

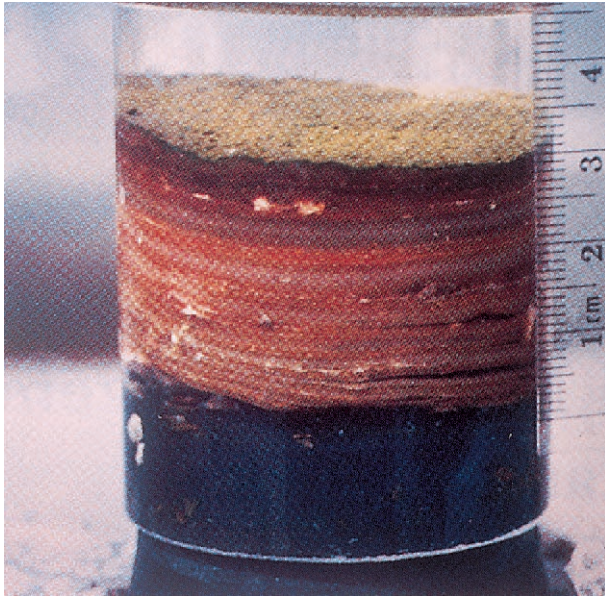


Figure 28.30 Microbial Mats. Microorganisms, through their metabolic activities, can create environmental gradients resulting in layered ecosystems. A vertical section of a hot spring (55°C) microbial mat, showing the various layers of microorganisms.

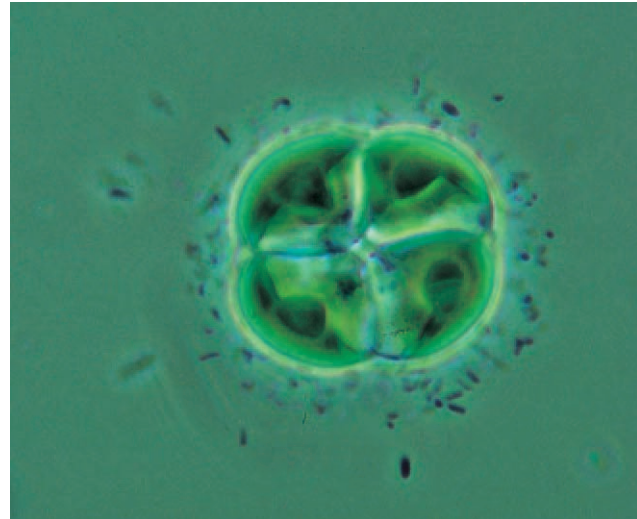


Figure 28.31 A Simple Ecosystem. An alga, which releases photosynthetically generated oxygen and organic matter to its environment, is surrounded by chemoheterotrophs that are using these products of primary production. These two types of microorganisms, producing and consuming oxygen and organic matter, form a self-regulating ecosystem ($\times 1,000$).

formation of terrestrial ecosystems prior to the development of vascular plants, and fossil microbial mats, called stromatolites, have been dated at over 3.5 billion years old (*see p. 423*). Molecular techniques and stable isotope measurements (see table 28.8) are being used to better understand these unique microbial communities.

1. What are biofilms? What types of surfaces on living organisms can provide a site for biofilm formation?
2. Why are biofilms important in human health?
3. What are microbial mats and where are they found?

Microorganisms and Ecosystems

Microorganisms, as they interact with each other and other organisms, and influence nutrient cycling in their specific microenvironments and niches, also contribute to the functioning of ecosystems. Ecosystems have been defined as “communities of organisms and their physical and chemical environments that function as self-regulating units.” These self-regulating biological units respond to environmental changes by modifying their structure and function.

Microorganisms in ecosystems can have two complementary roles: (1) the synthesis of new organic matter from CO_2 and other inorganic compounds during **primary production** and (2) decomposition of this accumulated organic matter. A simple self-regulating ecosystem in which primary production of organic

matter occurs is shown in **figure 28.31**. This consists of an alga and a “halo” of surrounding bacteria that are using the organic matter formed by algal photosynthesis as a carbon, electron, and energy source, and returning the organic matter to its original mineral constituents. Self-regulation in this ecological unit is shown by its response to light. Decreased light fluxes lead to a decrease in photosynthesis and organic matter release. Under these conditions, the heterotrophic bacterial community will be limited and its activity and biomass may be decreased.

The general relationships between the **primary producers** that synthesize organic matter, the heterotrophic **decomposers**, and the consumers are illustrated in **figure 28.32**. Microorganisms of different types contribute to each of these complementary relationships.

In terrestrial environments the primary producers are usually vascular plants. In freshwater and marine environments, the cyanobacteria and algae (*see section 21.3 and chapter 26*) play a similar role. The major energy source driving primary production is light in both habitats although in hydrothermal and hydrocarbon seep areas, chemotrophic ecosystems occur. The higher **consumers**, including humans, are chemoheterotrophs. These consumers depend on the “life support systems” provided by organisms that accumulate and decompose organic matter.

Microorganisms thus carry out many important functions as they interact in ecosystems, including:

1. Contributing to the formation of organic matter through photosynthetic and chemosynthetic processes.

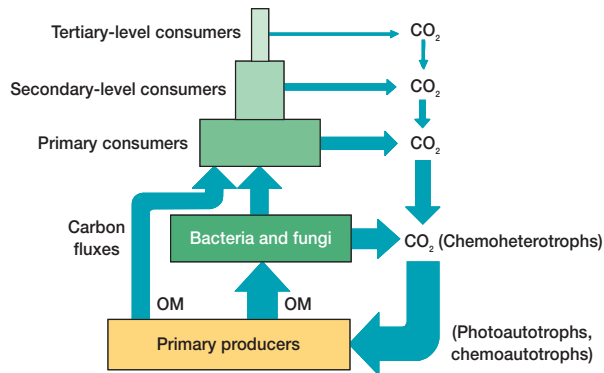


Figure 28.32 The Vital Role of Microorganisms in Ecosystems. Microorganisms play vital roles in ecosystems as primary producers, decomposers, and primary consumers. Carbon is fixed by the primary producers, including microorganisms, which use light or chemically bound energy. Chemoheterotrophic bacteria and fungi serve as the main decomposers of organic matter, making minerals again available for use by the primary producers. Ciliates and flagellates, important microbial primary consumers, feed on the bacteria and fungi, recycling nutrients as part of the microbial loop. Organic matter (OM).

- Decomposing organic matter, often with the release of inorganic compounds (e.g., CO_2 , NH_4^+ , CH_4 , H_2) in mineralization processes.
- Serving as a nutrient-rich food source for other chemoheterotrophic microorganisms, including protozoa and animals.
- Modifying substrates and nutrients used in symbiotic growth processes and interactions, thus contributing to biogeochemical cycling.
- Changing the amounts of materials in soluble and gaseous forms. This occurs either directly by metabolic processes or indirectly by modifying the environment.
- Producing inhibitory compounds that decrease microbial activity or limit the survival and functioning of plants and animals.
- Contributing to the functioning of plants and animals through positive and negative symbiotic interactions.

- Define the following terms: ecosystem, primary production, decomposer, mineralization.
- List important functions of higher consumers in natural environments.
- What are the important functions of microorganisms in ecosystems?

Microorganism Movement between Ecosystems

Microorganisms constantly are moving and being moved between ecosystems. This often happens naturally in many ways: (1) soil is transported around the Earth by windstorms and falls on land ar-

eas and waters far from its origins; (2) rivers transport eroded materials, sewage plant effluents, and urban wastes to the ocean; and (3) insects and animals release urine, feces, and other wastes to environments as they migrate around the Earth. When plants and animals die after moving to a new environment, they decompose and their specially adapted and coevolved microorganisms (and their nucleic acids) are released (see section 29.2). The fecal-oral route of disease transmission, often involving foods and waters, and the acquisition of diseases in hospitals (nosocomial infections) are important examples of pathogen movement between ecosystems. Each time a person coughs or sneezes, microorganisms also are being transported to new ecosystems.

Humans also both deliberately and unintentionally move microorganisms between different ecosystems. This occurs when microbes are added to environments to speed up microbially mediated degradation processes (see *bioremediation*, section 42.4) or when a plant-associated inoculum such as *Rhizobium*, is added to a soil to increase the formation of nitrogen-fixing nodules on legumes (see pp. 675–78). One of the most important accidental modes of microbial movement is the use of modern transport vehicles such as automobiles, trains, ships, and airplanes. These often rapidly move microorganisms long distances.

The fate of microorganisms placed in environments where they normally do not live, or of microorganisms returned to their original environments, is important both theoretically and practically. Pathogens that are normally associated with an animal host are greatly affected by such movement because these microorganisms largely have lost their ability to compete effectively with microorganisms indigenous to other environments. Upon moving to a new environment, the population of viable and culturable pathogens gradually decreases. However, more sensitive viability assessment procedures, particularly molecular techniques, indicate that **nonculturable microorganisms**, as observed with *Vibrio* (see section 6.5), may play critical roles in disease occurrence.

Many studies have been directed toward learning why microorganisms which have coevolved with animals gradually die after being released to soils and waters. Among the possibilities are predation by protozoa, *Bdellovibrio* (see pp. 510–12) and other organisms, lack of space, lack of nutrients, and the presence of toxic substances. After many years of study, it appears that the major reason “foreign” microorganisms die out is that they can no longer compete effectively with indigenous microorganisms for the low amounts of nutrients present in the environment.

Even microorganisms recovered from a particular environment, after growth in the laboratory on rich media, may lose their ability to survive when placed back in their original environment. The cause may be physical or physiological. From a physical standpoint the microorganisms may find themselves outside their protected physical niche, where they can be consumed by protozoa and other predators as noted previously. On the other hand, after growth in rich laboratory media, they may have lost the ability to compete physiologically with the native populations. It is of interest that these foreign microorganisms survive longer outside of their original hosts

Table 28.7 Characteristics of Extreme Environments in which Microorganisms Grow

Stress	Environmental Conditions	Microorganisms Observed
High temperature	110–113°C, deep marine trenches	<i>Pyrolobus fumarii</i> <i>Methanopyrus kandleri</i> <i>Pyrodictium abyssi</i>
	67–102°C, marine basins	<i>Pyrococcus abyssi</i>
	85°C, hot springs	<i>Thermus</i> <i>Sulfolobus</i> <i>Thermothrix thiopara</i>
Low temperature	75°C, sulfur hot springs	
	–12°C, antarctic ice	<i>Psychromonas ingrahamii</i>
Osmotic stress	13–15% NaCl	<i>Chlamydomonas</i>
	25% NaCl	<i>Halobacterium</i> <i>Halococcus</i>
Acidic pH	pH 3.0 or lower	<i>Saccharomyces</i> <i>Thiobacillus</i>
	pH 0.5	<i>Picrophilus oshimae</i>
	pH 0.0	<i>Ferroplasma acidarmanus</i>
Basic pH	pH 10.0 or above	<i>Bacillus</i>
Low water availability	$a_w = 0.6–0.65$	<i>Torulopsis</i> <i>Candida</i>
Temperature and low pH	85°C, pH 1.0	<i>Cyanidium</i> <i>Sulfolobus acidocaldarius</i>
Pressure	500–1,035 atm	<i>Colwellia hadaliensis</i>
Radiation	1.5 million rads	<i>Deinococcus radiodurans</i>

at lower temperatures (e.g., in polar regions, ice, or frozen foods). Under these conditions, the survival time of these foreign bacteria is greatly extended.

1. How can microorganisms move between different ecosystems?
2. Why might microorganisms isolated from soil or water, after being grown in the laboratory, lose the ability to survive in the environment from which they were taken?
3. What is the effect of temperature on the die-out rate of microorganisms that have been moved to a new, foreign environment?

Stress and Ecosystems

Microorganisms function in ecosystems that develop under a wide range of environmental conditions. These have varied pHs, temperatures, pressures, salinity, water availability, and ionizing radiation as summarized in **table 28.7**. Such stress factors have major effects on microbial populations and communities, and can create an **extreme environment**, as shown in **figure 28.33**. In these cases high salt concentrations, extreme temperature, and acidic conditions have affected the microbial communities. The microorganisms that survive in such envi-

ronments are described as **extremophiles**, and such extreme environments are usually considered to have decreased microbial diversity, as judged by the microorganisms that can be cultured. With the increased use of molecular detection techniques, however, it appears that there is surprising diversity among the microorganisms that cannot be cultured from these extreme environments. Further work to establish relationships between the microorganisms that can be observed and detected by these molecular techniques and culturable microorganisms will be required in the future. [The influence of environmental factors on growth \(pp. 121–31\)](#)

Many microbial genera have specific requirements for survival and functioning in such so-called extreme environments. For example, a high sodium ion concentration is required to maintain membrane integrity in many halophilic bacteria, including members of the genus *Halobacterium*. Halobacteria require a sodium ion concentration of at least 1.5 M, and about 3 to 4 M for optimum growth. [Halophilic archaea \(pp. 123; 461–63\)](#)

The bacteria found in deep-sea environments have different pressure requirements, depending on the depth from which they are recovered. These bacteria can be described as **baro-** or **piezo-tolerant bacteria** (growth from approximately 1 to 500 atm), **moderately barophilic bacteria** (growth optimum 5,000 meters, and still able to grow at 1 atm), and **extreme barophilic bacteria**, which require approximately 400 atm or higher for growth (*see chapter 29*).

Intriguing changes in basic physiological processes occur in microorganisms functioning under extreme acidic or alkaline conditions. These acidophilic and alkalophilic microorganisms have markedly different problems in maintaining a more neutral internal pH and chemiosmotic processes (*see chapter 9*). Obligately acidophilic microorganisms can grow at a pH of 3.0 or lower, and major pH differences can exist between the interior and exterior of the cell. These acidophiles include members of the genera *Thiobacillus*, *Sulfolobus*, and *Thermoplasma*. The higher relative internal pH is maintained by a net outward translocation of protons. This may occur as the result of unique membrane lipids, hydrogen ion removal during reduction of oxygen to water, or the pH-dependent characteristics of membrane-bound enzymes.

Recently, an archaeal iron-oxidizing acidophile, *Ferroplasma acidarmanus*, capable of growth at pH 0, has been isolated from a sulfide ore body in California. This unique prokaryote, capable of massive surface growth in flowing waters in the subsurface (**figure 28.34**), possesses a single peripheral cytoplasmic membrane and no cell wall.

The extreme alkalophilic microorganisms grow at pH values of 10.0 and higher and must maintain a net inward translocation of protons. These obligate alkalophiles cannot grow below a pH of 8.5 and are often members of the genus *Bacillus*; *Micrococcus* and *Exiguobacterium* representatives have also been reported. Some photosynthetic cyanobacteria also have similar characteristics. Increased internal proton concentrations may be maintained by means of coordinated hydrogen and sodium ion fluxes.



(a)



(b)



(c)

Figure 28.33 Microorganisms Growing in Extreme Environments. Many microorganisms are especially suited to survive in extreme environments. **(a)** Salterns turned red by halophilic algae and halobacteria. **(b)** A hot spring colored green and blue by cyanobacterial growth. **(c)** A source of acid drainage from a mine into a stream. The soil and water have turned red due to the presence of precipitated iron oxides caused by the activity of bacteria such as *Thiobacillus*.



Figure 28.34 Massive Growth of the Extreme Acidophile *Ferropasma acidarmanus* in a California Mine. Slime streamers of *Ferropasma acidarmanus*, an archaean, which have developed within pyritic sediments at and near pH 0. This unique procaryote has a single plasma membrane and no cell wall.

Box 28.2

The Potential of Microorganisms from High-Temperature Environments for Use in Modern Biotechnology

There is great interest in the characteristics of procaryotes isolated from the outflow mixing regions above deep hydrothermal vents that release water at 250 to 350°C. This is because these procaryotes can grow at temperatures as high as 113°C. The problems in growing these microorganisms, often archaea, are formidable. For example, to grow some of them, it will be necessary to use special culturing chambers and other specialized equipment to maintain water in the liquid state at these high temperatures.

Such microorganisms, termed hyperthermophiles, with optimum growth temperatures of 80°C or above (*see p. 126*), confront unique challenges in nutrient acquisition, metabolism, nucleic acid replication, and growth. Many of these are anaerobes that depend on elemental sul-

fur as an oxidant and reduce it to sulfide. Enzyme stability is critical. Some DNA polymerases are inherently stable at 140°C, whereas many other enzymes are stabilized in vivo with unique thermoprotectants. When these enzymes are separated from their protectant, they lose their unique thermostability.

These enzymes may have important applications in methane production, metal leaching and recovery, and for use in immobilized enzyme systems. In addition, the possibility of selective stereochemical modification of compounds normally not in solution at lower temperatures may provide new routes for directed chemical syntheses. This is an exciting and expanding area of the modern biological sciences to which microbiologists can make significant contributions.

Observations of microbial growth at temperatures approaching 113°C in thermal vent areas, or of **hyperthermophiles**, (**Box 28.2**; *see also Box 6.1*) indicate that this area will continue to be a fertile field for investigation. For some successful microorganisms, an extreme environment may not be “extreme” but required and even, perhaps, ideal. [Thermophilic microorganisms](#) (*pp. 126; 463*)

1. What are the main factors that lead to the creation of extreme environments?
2. Why are molecular techniques possibly changing our view of these environments?
3. What is unique about *Ferroplasma acidarmanus*?

28.5 Methods in Microbial Ecology

A wide variety of techniques can be used to evaluate the presence, types, and activities of microorganisms as populations, communities, and parts of ecosystems (**table 28.8**). Measurements made by these techniques can span a range of time scales and physical dimensions. In marine, freshwater, sewage, and plant root environments, as examples, responses can be measured in seconds and minutes. For deep marine and soil organic matter changes, a time scale of years, decades, or even centuries may be required. The physical scale used in a study may range from a single bacterium and its microenvironment to a lake, ocean, or an entire plant-soil system.

As noted at the beginning of this chapter, a fundamental problem in studying microorganisms in nature is the inability to culture and characterize most organisms that can be observed. This long-standing problem is now being approached by the use of molecular techniques, by which nonculturable microorganisms can be characterized and compared with known genomic sequences (*see chapter 19*).

Microbial community diversity can be assessed by several approaches, including molecular phylogeny based on analyses of small subunit (SSU) ribosomal RNA (*see pp. 433–35*). Small amounts of DNA also can be recovered from environmental samples or individual cells and “amplified” by use of the polymerase chain reaction (PCR). [The polymerase chain reaction](#) (*pp. 326–27*)

As noted in table 28.8, some of these techniques are limited in terms of the types of samples that can be analyzed. This may be due to low microbial populations (marine and some freshwater samples) or high concentrations of interfering organic matter or particulates in samples. In contrast, the newer molecular procedures, such as direct DNA extraction, DNA amplification fingerprinting (DAF), 16S and 18S RNA-based phylogenetic analysis, PCR, and DNA probe and hybridization techniques, are applicable to a wider variety of samples.

Recently gel array microchips containing mixtures of probes, called “genosensors,” or **microarrays**, have been developed (*see pp. 354; 1018*). These allow the detection of small subunit ribosomal RNA from mixed populations. In addition, probes can detect specific groups of microorganisms such as the iron- and manganese-oxidizing sheathed bacteria by the use of 16S rRNA-based probes.

Many newer and more sensitive procedures are now available, including the use of radioactive substrates and sophisticated techniques to measure the viability and activity of individual microorganisms. Hybridization techniques can be used to “probe” colonies or single cells to determine if they contain specific DNA or RNA sequences. The technique of whole-cell hybridization has progressed to the point that “subcluster-specific” probes have been developed. These allow the simultaneous detection of different microbial types in the same preparation (*see figure 29.9, p. 643*).

In most studies employing these molecular approaches to analyze complex microbial communities, nucleic acids have been extracted from the sample, followed typically by cloning and further genomic and phylogenetic analyses. The specific source of the nucleic acids that are being studied is not known. Because of

Table 28.8 Methods Used to Study Microorganisms in Different Environments^a

Characteristic Evaluated	Technique Employed or Property Measured	Environment				
		Marine	Freshwater	Sewage	Soil	Food
Nutrients	Chemical analysis (e.g., C, N, P)	++	++	++	++	++
	COD (chemical oxygen demand)	—	+	++	—	—
	BOD (biochemical oxygen demand)	—	+	++	—	—
Microbial biomass	Photosynthetic pigments	++	++	—	—	—
	Filtration and dry weight	++	++	+	—	+
	Measurement of chemical constituents (e.g., ATP, muramic acid, polybetahydroxybutyric [PHB] acid, lipopolysaccharides)	++	++	++	++	++
	Microscopy and biovolume conversion to biomass using conversion factors	++	++	+	++	++
	Fumigation incubation/extraction	—	—	+	++	—
Microbial numbers/types	Glucose-amended respiration	—	—	+	++	—
	Microscopic procedures—epifluorescence/phase contrast	++	++	++	++	+
	Flow cytometry	++	++	+	—	—
	Immersion/insertion slides and films	+	++	+	++	—
	Viable enumeration procedures (culture, microscopy)	++	++	++	++	++
	Direct microorganism isolation	++	++	++	++	++
	Thin sections of samples	—	—	—	++	++
	Scanning electron microscopy	++	++	++	++	++
	Direct DNA extraction and analysis	++	++	++	++	++
	Polymerase chain reaction (PCR) with species-specific primers	++	++	++	++	++
	DNA probes and hybridization techniques for DNA and RNA	++	++	++	++	++
	In situ PCR with 16S or 18S primers	++	++	++	++	++
Microbial viability and turnover	Micromanipulation and single-cell PCR/phylogenetic analysis	++	++	++	++	++
	Nalidixic acid and microscopic observation (inhibitor prevents cell division, resulting in elongated active cells); direct viable counts	++	++	++	++	++
	Bioluminescence assessment (green fluorescent protein)	+	+	+	+	+
Microbial activity	Stable and radioactive isotope studies	++	++	+	++	++
	Microscopy with reducible dyes	++	++	++	++	++
	Autoradiography	++	++	++	++	++
	Enzyme activity assays	+	+	++	++	+
	Microcalorimetry	—	—	+	+	+
	RNA-based function probes	++	++	++	++	++
	Gas exchange (O ₂ , CO ₂ , N ₂ , CH ₄)	++	++	++	++	—
	Assessment of fungal and bacterial contributions by use of selective antibiotic inhibition	+	+	—	++	—
Community structure	Substrate utilization rate	++	++	++	++	+
	Fluorescent substrate hydrolysis	++	++	++	++	+
	Microscopic analyses of diversity	++	++	++	++	+
	Physiological diversity of microbial isolates	++	++	++	++	+
	16S or 18S ribosomal RNA analysis	++	++	++	++	+
	(including SSCP [single-strand conformation polymorphism], DGGE [denaturing gradient gel electrophoresis], TGGE [temperature gradient gel electrophoresis], etc.)	++	++	++	++	+
	DNA amplification fingerprinting (DAF)	++	++	++	++	+

^aMajor uses are noted with two plus signs (++) and minor uses are noted with one plus (+); minimal or no use noted with (—).

improvements in the sensitivity of the polymerase chain reaction (see section 14.3), it is possible to study nucleic acids recovered from individual cells after isolating them with **optical tweezers** (a laser beam is used to drag a microbe away from its neighbors) or a **micromanipulator**. In the micromanipulator procedure, a desired cell or cellular organelle is drawn up into a fine capillary

after direct observation (**figure 28.35**). PCR amplification of nucleic acids from an isolated cell or organelle allows one to obtain sequence data for use in phylogenetic analysis (see chapter 19). For example, it has been possible to establish the phylogenetic relationship of a mycoplasma recovered from the flagellate *Koruga bonita* (**figure 28.36**).

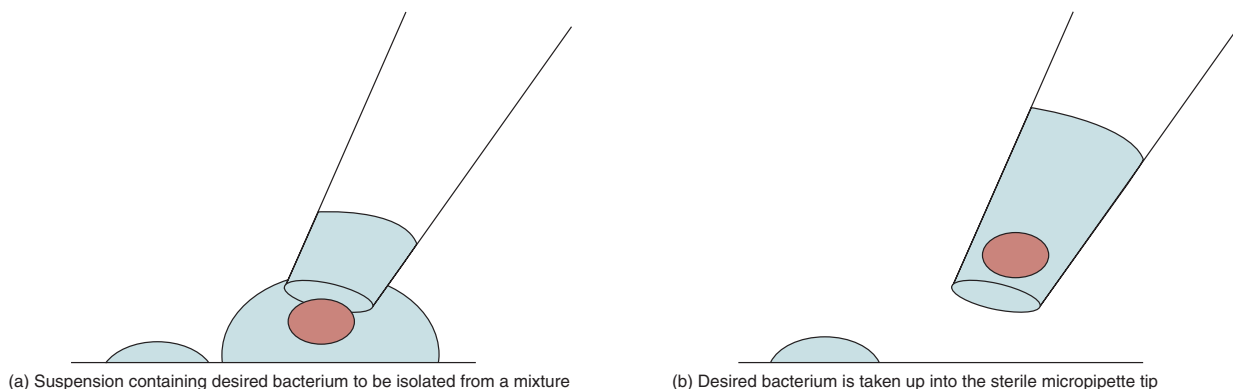


Figure 28.35 Recovery of Single Cells or Cell Organelles from Complex Natural Mixtures Using Micromanipulation. By use of an inverted phase-contrast microscope and a micromanipulator, a microorganism can be recovered for direct molecular analysis. **(a)** The bacterium to be isolated is placed below the micromanipulator tip (diameter 5 to 10 μm) and a slight vacuum is drawn. **(b)** The desired single bacterium is drawn up into the micropipette tube and is ready for molecular analysis.

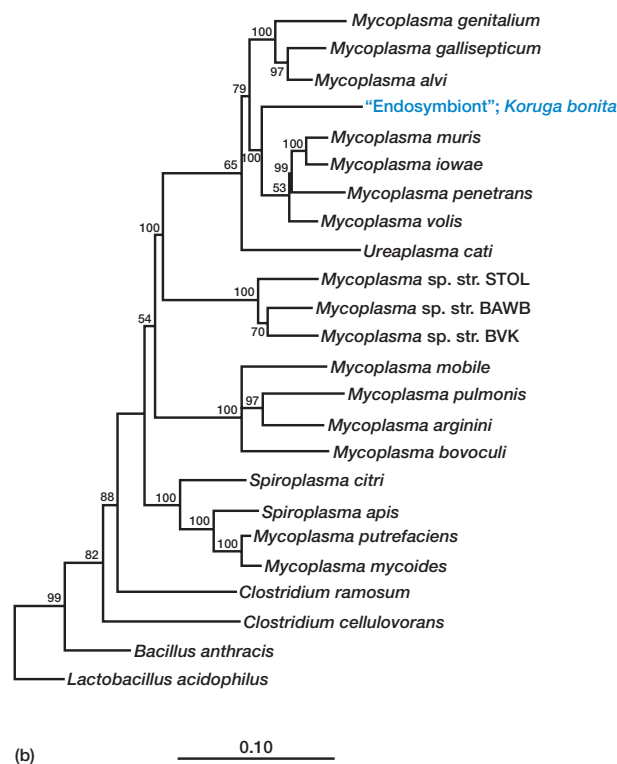
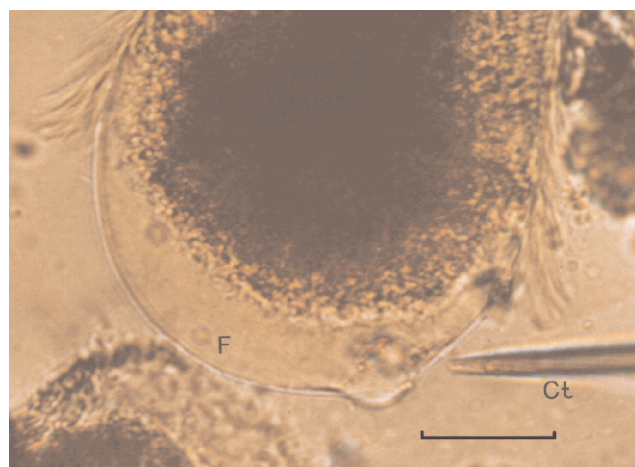


Figure 28.36 Combining Micromanipulation for Isolation of Single Cells or Organelles with the Polymerase Chain Reaction (PCR). **(a)** Recovery of an endosymbiotic mycoplasma from single cell of the flagellate *Koruga bonita* by micromanipulation (bar = 10 μM) and **(b)** phylogenetic analysis of the recovered mycoplasma following PCR amplification and sequencing of the PCR products, with the bar indicating 10% estimated sequence divergence. *Lactobacillus acidophilus* is the outgroup reference. This approach makes it possible to link a specific microorganism or organelle, isolated from a natural environment, to its molecular sequence and phylogenetic information. Flagellate (F), capillary tube (Ct).

With this single cell-based approach, it is now possible to link specific microbial structures observed under the microscope to the phylogenetic information contained in that cell or organelle. This should improve our understanding of the role of individual microorganisms in complex microbial assemblages.

Symbioses and microbial ecology involve complex relationships, the subtlety of which are only beginning to be understood. This involves not only finding out what microorganisms are there (the first formidable challenge) but also determining what the organisms are doing and how they are interacting at various temporal and spatial scales. These tasks are central to understanding mi-

crobial associations with other organisms and the environment, which is the essence of microbial ecology.

1. Why are “classic” microscopic and physical methods still used for the study of microorganisms when molecular techniques are available?
2. What are optical tweezers and micromanipulators?
3. What time scales can be used when studying the activity of microorganisms?
4. What is a genosensor or microarray system?

Summary

1. Most microorganisms that can be observed in complex natural assemblages under a microscope cannot be grown at the present time. Molecular techniques are making it possible to obtain information on these uncultured microorganisms.
2. Microbial ecology is the study of microorganisms' interactions with their living and nonliving environments. Symbiosis is a narrower term that means “together life,” the study of organism-organism interactions.
3. A microorganism functions in a physical location that can be described as its microenvironment. The resources available in a microenvironment and their time of use by a microorganism describe the niche. Pores are important microenvironments that can protect bacteria from predation.
4. One organism may grow on the surface of another organism as an ectosymbiont or inside another organism as an endosymbiont. Organisms may have other organisms on their surface and inside them at the same time, an ecto/endosymbiosis.
5. Positive interactions (**figure 28.1**) include mutualism (mutually beneficial and obligatory), proto cooperation (mutually beneficial, not obligatory), and commensalistic (product of one organism can be used beneficially by another organism). Negative interactions include predation (use and ingesting/killing a larger or smaller prey), parasitism (a longer-term internal maintenance of another organism or acellular infectious agent), and amensalism (a microbial product can inhibit another organism). Competition involves organisms competing for space or a limiting nutrient. The quality of these interactions can change, depending on the environment and the characteristics of the particular organisms.
6. Mutual advantage is central to positive organism-organism interactions. These interactions can be based on material transfers related to energetics, cell-to-cell communication, or physical protection. With several important mutualistic interactions, chemolithotrophic microorganisms play a critical role in making organic matter available for use by an associated organism (e.g., endosymbionts in *Riftia*).
7. Predation and parasitism are closely related. Predation has many positive effects on populations of predators and prey. These include the microbial loop (returning minerals immobilized in organic matter to mineral forms for reuse by chemotrophic and photosynthetic primary producers), protection of prey from heat and damaging chemicals, and possibly aiding pathogenicity, as with *Legionella*.
8. A consortium is a physical association of organisms that have a mutually beneficial relationship based on positive interactions.
9. Syntrophism simply means growth together. It does not require physical contact but only a mutually positive transfer of materials, such as interspecies hydrogen transfer.
10. The rumen is an excellent example of a mutualistic interaction between a ruminant and its complex microbial community. In this microbial community, complex plant materials are broken down to simple organic compounds that can be absorbed by the animal, as well as forming waste gases such as methane that are released to the environment (**figure 28.8**).
11. Proto cooperative interactions are beneficial for both organisms but are not obligatory (**figure 28.9**). Important examples are marine animals, including *Alvinella*, *Rimicarus*, and *Eubosrichus*, that involve interactions with hydrogen sulfide-oxidizing chemotrophs.
12. These varied positive and negative interactions occur in complex biological systems and lead to feedback responses of varied members of a biological community.
13. Microbes can be present as individual cells, as populations of similar organisms, or as mixtures of populations or communities. These populations and communities can be parts of self-regulating ecosystems.
14. Microorganisms—functioning with plants, animals, and the environment—play important roles in nutrient cycling, which is also termed biogeochemical cycling. Assimilatory processes involve incorporation of nutrients into the organism's biomass during metabolism; dissimilatory processes, in comparison, involve the release of nutrients to the environment after metabolism (**figure 28.18**).
15. Biogeochemical cycling involves oxidation and reduction processes, and changes in the concentrations of gaseous cycle components, such as carbon, nitrogen, and sulfur can result from microbial activity.
16. Microorganisms serve as primary producers that accumulate organic matter (**figure 28.32**). Energy sources include hydrogen, sulfide, and methane. In addition, many chemoheterotrophs decompose the organic matter that primary producers accumulate and carry out mineralization, the release of inorganic nutrients from organic matter.
17. Major organic compounds used by microorganisms differ in structure, linkage, elemental composition, and susceptibility to degradation under aerobic and anaerobic conditions. Lignin is degraded only under aerobic conditions, a fact that has important implications in terms of carbon retention in the biosphere.
18. In terms of effects on humans, metals can be considered in three broad groups: (1) the noble metals, which have antimicrobial properties but which do not have negative effects on humans; (2) metals such as mercury and lead, from which toxic organometallic compounds can be formed; and (3) certain other metals, which are antimicrobial in ionic form, such as copper and zinc. The second of these groups is of particular concern.
19. Biofilms, or layers of microorganisms, are widespread and are formed on a wide variety of living and nonliving surfaces (**figure 28.28**). These are important in disease occurrence and the survival of pathogens. Biofilms can develop to form complex layered ecosystems.
20. Most disease-causing bacteria from the intestinal tract of humans and other higher organisms do not survive in the environment. At lower temperatures, however, increased survival can occur.

21. Decreased species diversity usually occurs in extreme environments, and many microorganisms that can function in such habitats, called extremophiles, have specialized growth requirements. For them, extreme environments can be required environments.
22. Many approaches can be used to study microorganisms in the environment (**table 28.8**). These include nutrient cycling, biomass,

numbers, activity, and community structure analyses. It is now possible to study the genetic characteristics of microorganisms that cannot be grown in the laboratory.

23. Methods presently being used make it possible to study presence, types, and activities of microorganisms in their natural environments (including soils, waters, plants, and animals). Although most microorganisms that can be observed still cannot be grown, molecular

techniques are making it possible to obtain information on these noncultured microorganisms.

24. Optical tweezers and micromanipulators can be used to recover individual cells or cell organelles from complex microbial communities. This makes it possible to obtain genomic and phylogenetic information from specific individual microbial cells for use in studies of microbial ecology (**figure 28.36**).

Key Terms

aerobic anoxygenic photosynthesis 614
amensalism 609
anammox process 616
assimilatory reduction 614
barotolerant or piezotolerant bacteria 624
biocontrol processes 609
biofilm 620
biogeochemical cycling 611
biomagnification 618
commensal 606
commensalism 606
community 596
competition 609
competitive exclusion principle 609
consortium 596
consumer 622
decomposer 622
denitrification 616
dissimilatory reduction 614
ecosystem 596
ecto/endosymbiosis 596

ectosymbiont 596
endosymbiont 596
extreme barophilic bacteria 624
extreme environment 624
extremophile 624
heterotrophic nitrification 615
hyperthermophile 626
immobilization 613
interspecies hydrogen transfer 604
lichen 598
magneto-aerotactic bacteria 616
microarray 626
microbial ecology 596
microbial loop 608
microbial mat 621
microenvironment 619
micromanipulator 627
mineralization 613
moderately barophilic bacteria 624
mutualism 598
mutualist 598

mycobiont 598
niche 619
nitrification 615
nitrogen fixation 616
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Questions for Thought and Review

1. It is evident that some microorganisms have a wide range of electron donors and acceptors that they can use, and others use only single oxidants and reductants. What are the advantages and disadvantages of each strategy?
2. Why might microorganisms prefer to grow in association with other microorganisms, as in biofilms, when they can have better access to nutrients as single cells?
3. Why is mutualism in association with multicellular organisms a major strategy of microorganisms? Why have these especially developed with sulfide and simple hydrocarbon-based systems?
4. What types of unique information do microscopic and molecular techniques provide and not provide?
5. How might you show that a microorganism found in a particular extreme environment is actually growing there?
6. Why is predation such a very important part of microbial ecology? Can a predator ever completely eliminate all of its prey?

7. How might you attempt to grow a microorganism in the laboratory to increase its chances of being a strong competitor when placed back in a natural habitat?
8. Considering the possibility of microorganisms functioning at temperatures approaching 120°C, what do you think the limiting factor for microbial growth at higher temperatures will be and why?
9. Where in their bodies do most people have noble metals? Why have these been used with more success than materials such as ceramics, which have been tested over many decades?
10. Compare the degradation of lignin and cellulose by microorganisms. What different environmental factors are required for these important polymers to be degraded?
11. Considering the intensive searches for unique microorganisms that have been carried out all over the world, where can we look for new microbes?

Critical Thinking Questions

1. Compare and contrast diversity among microorganisms with diversity among macroorganisms.
2. Describe a naturally occurring niche on this planet that you believe is inhospitable to microbial life. Explain, in light of what is known about extremophiles, why you believe this environment will not support microbial life.

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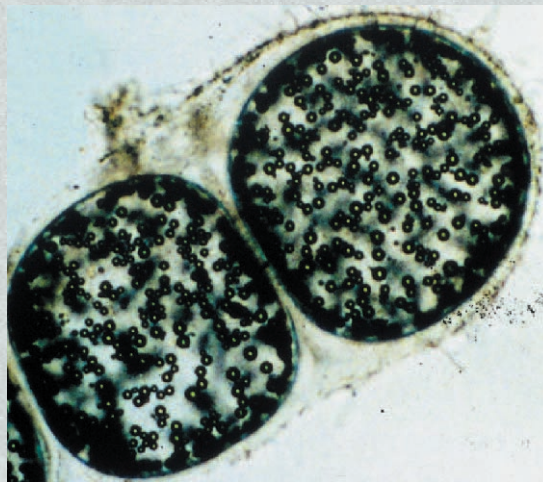
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CHAPTER 29

Microorganisms in Aquatic Environments



New procaryotes are discovered at locations where reduced and oxidized nutrients mix. This giant bacterium, *Thiomargarita namibiensis*, about 100 to 300 μm in diameter, accumulates sulfur from sediments in its refractive sulfur granules and nitrate from overlying waters to support its growth. *Thiomargarita*, which resembles a string of pearls, is found off the coast of Namibia.

Outline

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Concepts

1. Oxygen diffuses through water at a slow rate, in comparison with the rate at which it diffuses through air; this is a defining characteristic of water as an environment for microbial growth and functioning.
2. Oxygen, once it is dissolved in water, can be used by microorganisms at a faster rate than it can be replenished. This leads to the creation of anaerobic zones. If light penetrates into these anaerobic zones, unique groups of photosynthetic microorganisms can arise. Specialized microbial communities also develop at the interface between anoxic and oxic regions.
3. Aquatic environments, both marine and freshwater, include vast volumes of cold water and ice. In addition to fresh water stored in glaciers and the polar regions, sea ice covers approximately 7% of the world's surface. Together with the cold (2 to 3°C), high pressure waters of the Earth's oceans, these cold environments are important sites of microbial survival and functioning.
4. Aquatic environments allow development of many unique types of microorganisms. These include microbes at deep hydrothermal vents and hydrocarbon seeps, and other microorganisms that exploit surfaces by gliding motility and attachment. Other microorganisms link spatially separated resources such as nitrate and sulfide. New microorganisms continue to be discovered in aquatic environments.
5. The biogeochemical cycles of carbon, nitrogen, sulfur, and phosphorus for lakes and marine environments involve interactions over both local and vast distances.
6. Phytoplankton (photosynthetic procaryotic and eucaryotic organisms) form the basis of primary production in most marine and freshwater environments. The phytoplankton release dissolved and particulate organic matter, which is used by heterotrophic bacteria, a portion of which are consumed by predators, releasing materials that are recycled for use by the phytoplankton, creating a microbial loop. Iron and nitrogen can limit these activities in different marine environments.
7. Disease-causing microorganisms, as well as organic materials, are constantly being added to waters. These can be transported over large distances, especially in rapidly moving rivers, lakes, and marine environments. Atmospheric-borne dusts and other materials, including pollutants, can be carried to the farthest regions of the oceans, freshwater bodies, and ice-covered regions of the world.
8. Many important human pathogens, such as *Shigella*, *Vibrio*, and *Legionella*, are found in waters. These can occur normally or can survive for various intervals after addition to waters. Often protozoa provide protection for such pathogens, especially when the protozoa are associated with biofilms.
9. Aquatic environments can serve as reservoirs and transmission routes for disease-causing microorganisms. A major goal of aquatic system management is to control pathogen survival and transfer.
10. Water is an important reservoir for the survival and dissemination of protozoa and viruses. These cannot be reliably controlled by chlorination. Protozoan pathogens include *Cyclospora* and *Giardia*.
11. The biological use of organic wastes follows regular and predictable sequences. Once these sequences are understood, more efficient sewage treatment systems can be created.
12. Sewage treatment can be carried out using large vessels where mixing and aeration can be controlled (conventional treatment). Constructed wetlands, where aquatic plants and their associated microorganisms are used, now are finding widespread applications in the treatment of liquid wastes.
13. Indicator organisms, which usually die off at slower rates than many disease-causing microorganisms, can be used to evaluate the microbiological quality of water.
14. Groundwater is an important source of drinking water, especially in suburban and rural areas. In too many cases, this resource is being contaminated by disease-causing microorganisms, especially from septic tanks.

Of all the water found on Earth, 97% is marine. Most of this water is at a temperature of 2 to 3°C and devoid of light; 62% is under high pressure (>100 atm). Microscopic phytoplankton and associated bacteria create a complex food web that can extend over long distances and extreme depths. The marine environment seems so vast that it will not be able to be affected by pollution; however, in coastal areas human activities are increasingly disrupting microbial processes and damaging water quality.

Fresh waters, although a small part of the waters on Earth, are extremely important as a source of drinking water. In many locations the contamination of surface and subsurface waters by domestic and industrial wastes causes environmental problems.

Marine and freshwater environments create unique niches for many specialized microorganisms needing habitats with a continuous water phase.

29.1 Aquatic Environments and Microorganisms

A major factor in aquatic environments is the movement of materials, whether they be gaseous, solid, or in the dissolved phase. These changes in concentration are part of the world of aquatic microorganisms; these microorganisms are able to respond rapidly to select their most suitable environment.

The mixing and movement of nutrients, O₂, and waste products that occur in freshwater and marine environments are the dominant factors controlling the microbial community. For example, in deep lakes or oceans, organic matter from the surface can sink to great depths, creating nutrient-rich zones where decomposition takes place. Gases and soluble wastes produced by microorganisms in these deep marine zones can move into overlying waters and stimulate the activity of other microbial groups. Similar processes take place on a lesser scale in nutrient-rich lakes, biofilms, and in microbial mats (*see pp. 620–22*) where gradients are established on a scale of micrometers.

Aquatic environments have varied surface areas and volumes. They are found in locations as diverse as the human body; drinks and beverages; and the usual places one would expect—rivers, lakes, and the oceans. They also occur in water-saturated zones in materials we usually describe as soils! These environments can range from alkaline to extremely acidic. The temperatures within which microorganisms function in aquatic environments can range from –5 to –15°C at the lower range, to at least 113°C in geothermal areas. Some of the most intriguing microbes have come from the study of high-temperature environments, including the now-classic studies of T. D. Brock and his coworkers at Yellowstone National Park which led to the discovery of *Thermus aquaticus*, the source of *Taq* polymerase (*see p. 326*). Hyperthermophilic microorganisms, including *Pyrolobus fumarii*, also have been isolated from hydrothermal vents in deep marine environments.

A goal of microbiologists is to isolate and culture unique marine microorganisms, especially in the search to find new antibiotic-producing microorganisms (**Box 29.1**). New techniques to collect microorganisms without changing temperature or pres-

Water is a very good servant, but it is a cruel master.

—John Bullen

Box 29.1

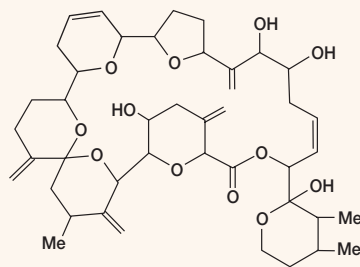
New Agents in Medicine: The Sea as the New Frontier

Most currently available antibiotics have been derived from soil microorganisms, primarily from the actinomycetes, but also from a lesser number of nonfilamentous gram-positive bacteria and fungi (see section 35.2). Over 100 products are in use as antibiotics, antitumor agents, and agrochemicals.

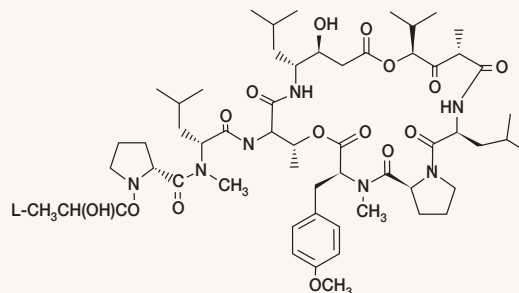
In recent years, with the need for additional compounds for use in medicine, marine microorganisms are receiving increased attention. Some of the newer chemicals that have been discovered are microalgal metabolites. There also is an interest in the culture of symbiotic marine microorganisms, including *Prochloron*, which are associated with macroscopic hosts. A variety of interesting compounds of unknown origin have been discovered. Many are assumed to be of microbial ori-

gin, but more work will be needed to establish this. Many biologists feel that marine microorganisms may provide unique bioactive compounds, including marine toxins, which do not occur in terrestrial microorganisms. There is a worldwide effort to better characterize the marine microbial community and to harness these often poorly studied organisms for modern medicine. Often the first challenge is to culture these organisms.

Examples of recent discoveries include goniiodomin A, a macrolide antifungal agent isolated from *Goniiodoma* (*Gonyaulax*) sp. (see **Box figure**, part a) and didemnin B, a strong antiviral and antitumor agent (see **Box figure**, part b). This latter compound is produced by members of the genus *Prochloron*.



(a)



(b)

New Chemotherapeutic Agents from the Sea. (a) Goniiodomin A, a macrolide antifungal agent. (b) Didemnin B, an antiviral and antitumor agent.

sure are now available. For example, Japanese and American scientists have constructed equipment to culture microorganisms that grow at 1,000 atm without decompressing them.

Gases and Aquatic Microorganisms

In aquatic environments the distance (on a microbial scale) from an air bubble or the water surface limits oxygen diffusion. Thus aquatic environments are termed **low oxygen diffusion environments**. The flux rate for oxygen through water is approximately 1/4,000 of that which would occur if the microorganisms were in direct association with a gas phase (**figure 29.1**).

Oxygen not only diffuses slowly through waters, its solubility is further decreased at higher temperatures and with lower pressure (**table 29.1**). Because of limited solubility and the low oxygen diffusion in waters, oxygen can be used by aerobic microorganisms faster than it can be replenished. This frequently leads to the formation of **hypoxic** or **anoxic** zones in aquatic environments. These zones allow specialized anaerobic microbes, both chemotrophic and phototrophic, to grow in the lower regions

of lakes where light can penetrate. In contrast, if the microorganisms are functioning in an extremely thin water film and oxygen-containing air is close to them, they are in **high oxygen diffusion environments**. Examples include soils, which will be discussed in chapter 30.

Low oxygen diffusion environments, which will be discussed in this chapter, can range from an ice cube in a cocktail glass to Arctic and Antarctic ice sheets (the latter make up 7% of the world's surface) and the vastness of marine and freshwater systems.

The second major gas in water, CO₂, plays many important roles in chemical and biological processes. The carbon dioxide-bicarbonate-carbonate equilibrium can control the pH in weakly buffered waters, or it can be controlled by the pH of strongly buffered waters. As shown in **figure 29.2**, the pH of distilled water, which is not buffered, is determined by the dissolved CO₂ in equilibrium with the air, and is approximately 5.0 to 5.5. In comparison, water strongly buffered at pH 8.0 contains CO₂ absorbed from the air, which is present primarily as bicarbonate (HCO₃⁻). When autotrophic microorganisms such as algae use CO₂, the pH of many waters will be increased.

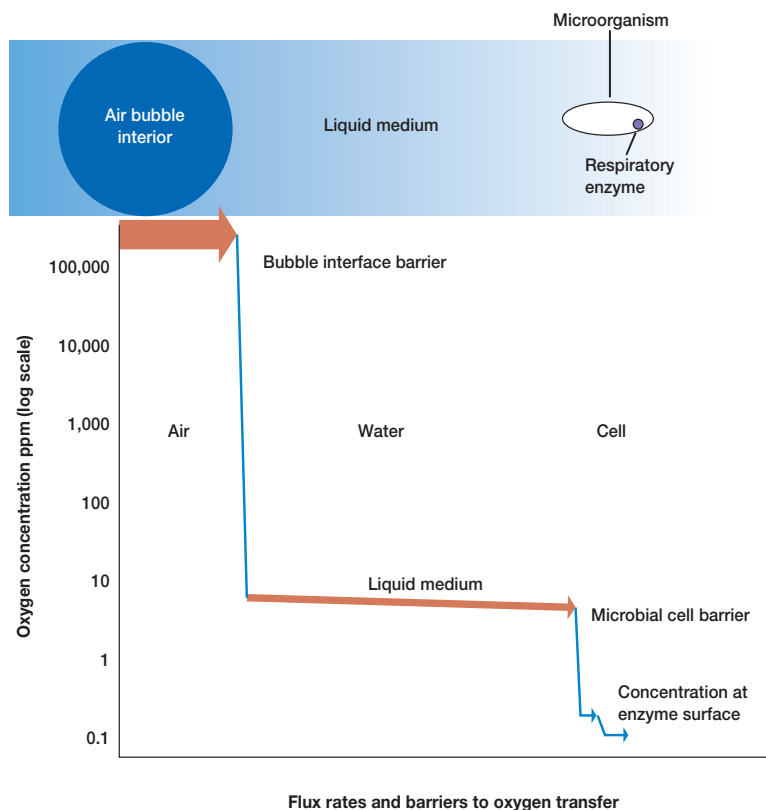


Figure 29.1 Oxygen Transfer in Water Is Limited. Oxygen diffuses at a high rate through air and at approximately 1/4,000 this rate through water. The wide arrow size represents the rapid oxygen flux rate through the air; the decrease in the intensity of the blue color indicates the oxygen concentration. After rapid diffusion through air, oxygen crosses the air/water barrier and then diffuses at a lower rate (the narrow arrow) and lower concentration through the water to the point at which it is used by a microorganism. The oxygen concentration (ppm) is expressed on a logarithmic scale, and the oxygen present at an enzyme can be less than one-millionth of that occurring in the air.

Table 29.1 The Effects of Water Temperature and Elevation on Dissolved O_2 Levels (mg/liter) in Water

Temperature ($^{\circ}C$)	Elevation above Sea Level (Meters)			
	0	1,000	2,000	3,000
0	14.6	12.9	11.4	10.2
5	12.8	11.2	9.9	8.9
10	11.3	9.9	8.8	7.9
15	10.0	8.9	7.9	7.1
25	9.1	8.1	7.1	6.4
30	8.2	7.3	6.4	5.8
35	7.5	6.6	5.9	5.3
40	6.9	6.1	5.4	4.9

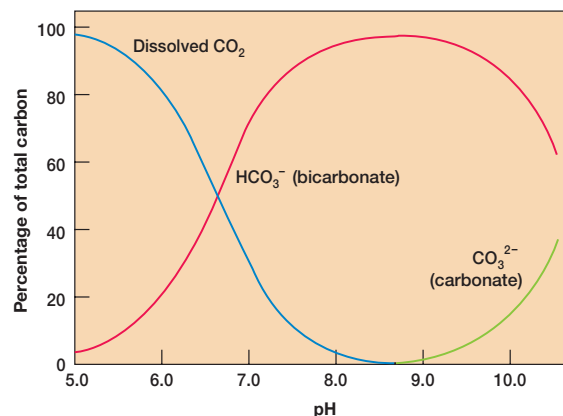


Figure 29.2 The Relationship of pH to Dissolved CO_2 . Atmospheric gases can affect the physical characteristics of water. The pH of water is influenced by the amount of dissolved carbon dioxide in the water, and also by the equilibrium of the dissolved carbon dioxide with bicarbonate and carbonate ions.

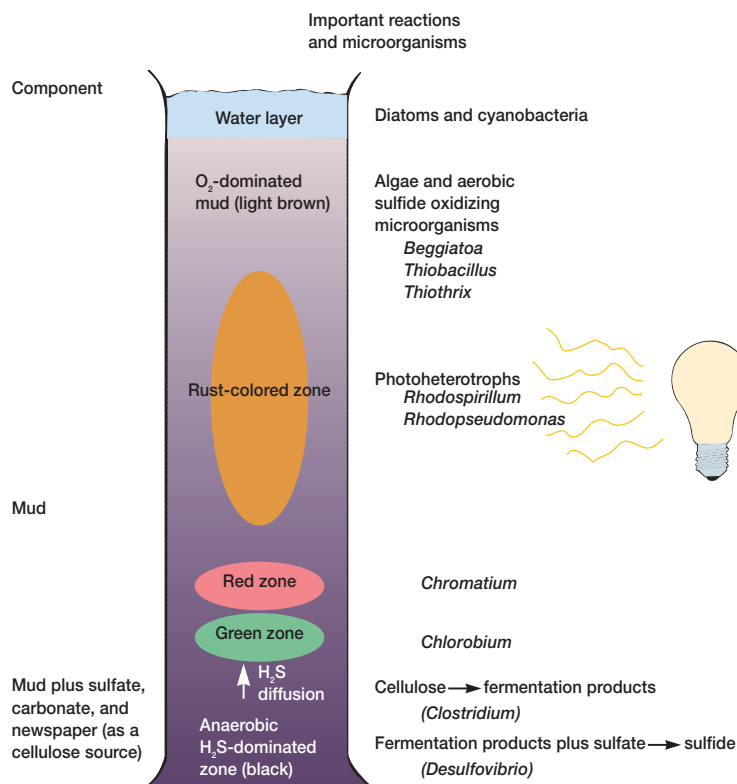


Figure 29.3 The Winogradsky Column. A microcosm in which microorganisms and nutrients interact over a vertical gradient. Fermentation products and sulfide migrate up from the reduced lower zone, and oxygen penetrates from the surface. This creates conditions similar to those in a lake with nutrient-rich sediments. Light is provided to stimulate the penetration of sunlight into the anaerobic lower region, which allows photosynthetic microorganisms to develop.

Other gases also are important in aquatic environments. These include nitrogen gas, used as a nitrogen source by nitrogen fixers; hydrogen, which is both a waste product and a vital substrate; and methane (CH₄). These gases vary in their water solubility, and methane is the least soluble of the three. Methane thus is an example of an ideal microbial waste product: once it is produced under anaerobic conditions, it leaves the microorganism's environment by diffusing up in the water column and being released to the atmosphere. This eliminates the problem of toxic waste accumulation that occurs with many microbial metabolic products, such as organic acids and ammonium ion.

1. What is an important defining characteristic of an aquatic environment?
2. Why is it critical to consider not only concentration but also flux rates of gases in waters?
3. How are anoxic zones created in aquatic environments?
4. How much oxygen, in milligrams per liter, is dissolved in water at normal room temperature and pressure? What are the effects of higher temperature and decreased pressure on oxygen solubility?
5. Why is carbon dioxide (CO₂) such a critical gas in aquatic environments? Consider both chemical equilibria and microbial processes.

Nutrients in Aquatic Environments

Nutrient concentrations in aquatic environments can vary from extremely low, in the range of micrograms of organic matter per liter, to levels approaching those in laboratory culture media, creating gradients that microorganisms exploit. High nutrient levels are found in polluted environments and sewage treatment plants, for example. With changes in nutrient levels, shifts between low-nutrient responsive and high-nutrient responsive microorganisms can occur. Nutrient turnover rates also vary. In marine environments the turnover time for nutrient processing may range from hundreds to thousands of years. In contrast, marsh and estuarine areas may have rapid rates of nutrient turnover, and a complex, diverse microbial community of rapidly responding microorganisms is present.

As noted in chapter 28, chemotrophically based marine and freshwater ecosystems are important recent discoveries, whether these occur in deep "black smoker" areas (*see pp. 126, 599–601*), in subsurface cave systems, or in shallower methane seep areas.

The **Winogradsky column**, usually constructed using a glass graduated cylinder, illustrates many interactions and gradients that occur in aquatic environments (**figure 29.3**). In this glass cylinder a layer of reduced mud is mixed with sodium sulfate, sodium carbonate, and shredded newspaper—a cellulose source—and additional

mud and water are placed in the column, which is then incubated in the light. A series of reactions occurs as the column begins to mature, with particular members of the microbial community developing in specific microenvironments in response to chemical gradients.

In the bottom of the column, cellulose is degraded to fermentation products by the genus *Clostridium* (see section 23.3). With these fermentation products available as reductants and using sulfate as an oxidant, *Desulfovibrio* produces hydrogen sulfide (see section 22.4). The hydrogen sulfide diffuses upward toward the oxygenated zone, creating a stable hydrogen sulfide gradient. In this gradient the photoautotrophs *Chlorobium* and *Chromatium* develop as visible olive green and purple zones. These microorganisms use hydrogen sulfide as an electron source, and CO₂, from sodium carbonate, as a carbon source. Above this region the purple nonsulfur bacteria of the genera *Rhodospirillum* and *Rhodopseudomonas* can grow. These photoheterotrophs use organic matter as an electron donor under anaerobic conditions and function in a zone where the sulfide level is lower. Both O₂ and hydrogen sulfide may be present higher in the column, allowing specially adapted microorganisms to function. These include *Beggiatoa* and *Thiothrix*, which use reduced sulfur compounds as a reductant and O₂ as an oxidant. In the upper portion of the column, diatoms and cyanobacteria may be visible. Green and purple photosynthetic bacteria (pp. 468–71, 487–88, 498–501); Cyanobacteria (pp. 471–76)

These commensalistic microorganisms, which develop sequentially, are dependent on the reductant originally provided as the cellulose or plant materials. When this reductant is exhausted, the column gradually becomes oxidized and the sulfide-dependent photosynthetic microorganisms and other anaerobes can no longer maintain themselves in the microcosm.

1. What are the sources of energy that support microbially based ecosystems in freshwater and marine environments?
2. What are the reasons for adding cellulose, sodium sulfate, and sodium carbonate to the Winogradsky column? Discuss this in relation to microbial groups that respond to these materials or their products.
3. What major microbial genera occur in the bottom of the Winogradsky column?

Nutrient Cycles in Aquatic Environments

The major source of organic matter in illuminated surface waters is photosynthetic activity, primarily from **phytoplankton** [Greek *phyto*, plant and *planktos*, wandering]. A common planktonic genus is *Synechococcus*, which can reach densities of 10⁴ to 10⁵ cells per milliliter at the ocean surface. Picocyanobacteria (very small cyanobacteria) may represent 20 to 80% of the total phytoplankton biomass upon which grazers depend.

As they grow and fix carbon dioxide to form organic matter, the phytoplankton acquire needed nitrogen and phosphorus

from the surrounding water. The nutrient composition of the water affects the final carbon-nitrogen-phosphorus (C:N:P) ratio of the phytoplankton, which is termed the **Redfield ratio**, named for the aquatic biologist A. C. Redfield. A commonly used value for this ratio is 106 parts C, 16 parts N, and 1 part P. This ratio is important for following nutrient dynamics, especially mineralization and immobilization processes, and for studying factors that limit microbial growth, especially the sensitivity of oceanic photosynthesis to atmospheric additions of nitrogen, sulfur, and iron.

Once the phytoplankton have grown, much of the organic matter fixed by these minute photosynthetic organisms then enters the **microbial loop** (figure 29.4). In the microbial loop, organic matter is recycled to carbon dioxide and minerals. Dissolved organic matter (DOM) released by the phytoplankton is used by the heterotrophic bacteria. The DOM is transformed to bacteria, which become part of the particulate organic matter (POM) pool.

These bacteria are then consumed and digested by a series of increasingly larger predators, including protozoa and metazoan zooplankters, releasing the carbon as CO₂ and the other nutrients in mineral forms to be cycled through the phytoplankton again. This results in the rapid cycling of nutrients at a stage between the primary producers and the higher consumers such as fish, leading to a decrease in ecosystem productivity. It has been suggested that this loss of carbon through the microbial loop is relatively greater in low-nutrient (oligotrophic) than in higher-nutrient (**copiotrophic**) environments.

It is important to note that the microbial loop operates best in aerobic environments where both photosynthetic microorganisms are active and the top consumers can function. If too much organic matter is added to a water, an anaerobic foul-smelling body of water is created that will not support top consumers such as fish. Once a body of water reaches this point, only major remediation efforts to limit nutrient inputs will restore it to its original condition and allow fish and other oxygen-requiring aquatic animals to survive.

Confined animal agriculture, especially when carried out near estuarine and coastal areas, can result in massive inputs of organic matter to waters that affect aquatic oxygen levels and the functioning of the microbial loop. One pig produces wastes equivalent to three to four people, and millions of tons of manure are produced per year in various confined beef cattle, hog, and poultry operations in the United States and elsewhere in the world.

1. What does the term phytoplankton mean?
2. Describe the Redfield ratio and its use.
3. What is the microbial loop? What role do protozoans play in this “loop”?
4. Define the terms oligotrophic and copiotrophic.

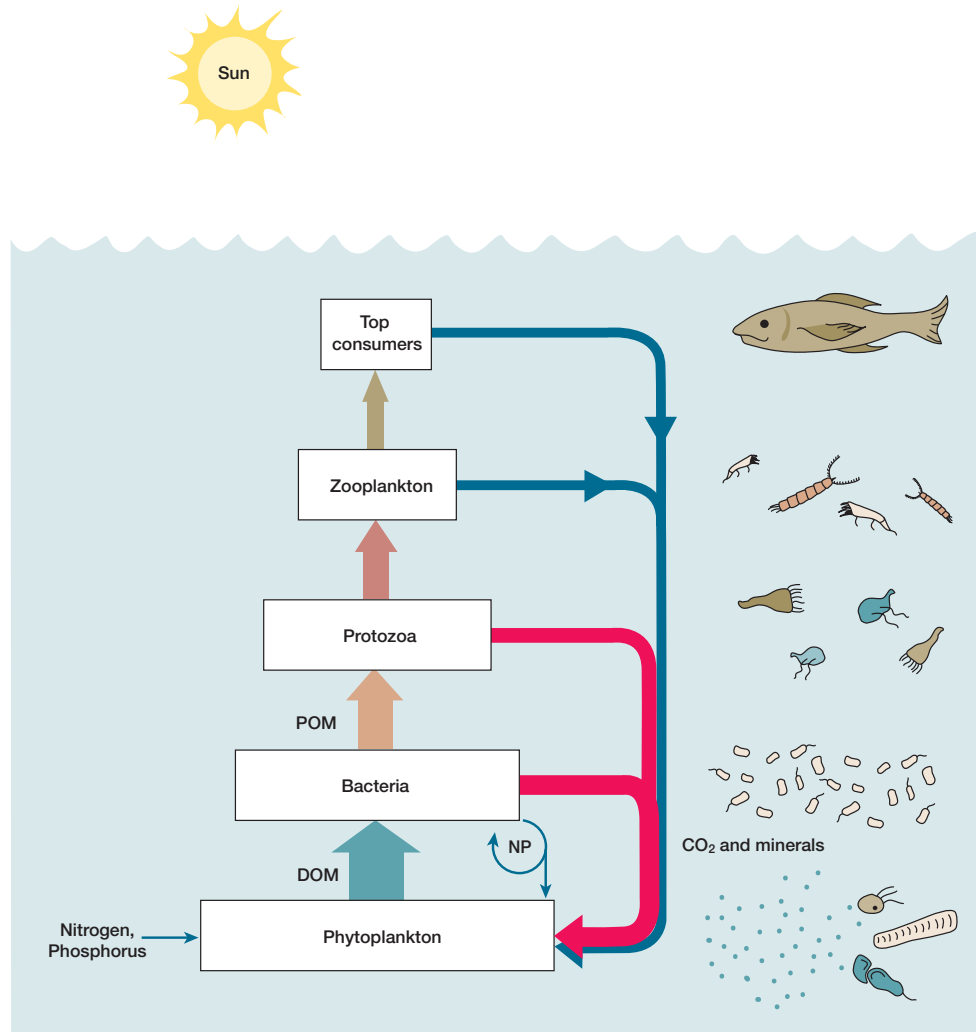


Figure 29.4 The Microbial Loop. A simplified view of the microbial loop, as it occurs in waters, is noted in red. In this loop a large portion of the organic matter synthesized by the phytoplankton during photosynthesis is released as dissolved organic matter (DOM). This DOM is used by the bacteria, which become part of the particulate organic matter (POM) pool. A portion of these bacteria are then used as a food source by protozoa. After digestion by the protozoa, a portion of the nutrients contained in the bacteria and protozoa is mineralized and “looped back” to the phytoplankton, making it unavailable to the higher trophic levels of the ecosystem. Arrow sizes representing the relative materials flows and organisms types (not to scale) are shown. Nitrogen (N), phosphorus (P).

29.2 The Microbial Community

Water provides an environment for a wide variety of microorganisms to survive and function (table 29.2). Microbial diversity depends on available nutrients, their varied concentrations (ranging from extremely low to very high levels), the transitions from aerobic to anaerobic zones, and the mixing of oxidants and reduc-

tants in this dynamic environment. In addition, the penetration of light into many anaerobic zones creates environments for certain types of photosynthetic microorganisms. Although important microorganisms found in aquatic environments have been discussed in the previous chapters on microbial diversity, it is important to discuss specific adaptations of microbes to these particular aquatic environments.

Table 29.2 Important Procaryotic Genera Found Primarily in Marine and Freshwater Environments

Group	Genera
Photoautotrophs	<i>Chlorobium</i>
	<i>Chloroherpeton</i>
	<i>Chromatium</i>
	<i>Pelodictyon</i>
	<i>Thiodictyon</i>
	<i>Thiopedia</i>
Photoheterotrophs	<i>Chloroflexus</i>
	<i>Heliobacterium</i>
	<i>Heliobacterium</i>
	<i>Rhodocyclus</i>
	<i>Rhodomicrobium</i>
	<i>Rhodopseudomonas</i>
	<i>Rhodospirillum</i>
Chemoheterotrophs	<i>Blastobacter</i>
	<i>Caulobacter</i>
	<i>Flexibacter</i>
	<i>Flexithrix</i>
	<i>Gemmobacter</i>
	<i>Hyphomicrobium</i>
	<i>Leucothrix</i>
	<i>Sphaerotilus</i>
Chemolithoautotrophs	<i>Beggiatoa</i> ^b
	<i>Gallionella</i>
	<i>Thioploca</i> ^b
	<i>Thiothrix</i> ^b
	<i>Thiovulum</i>

Sources: Data from V. M. Gorlenko, et al., "The Ecology of Aquatic Micro-Organisms," *Die Binnengewässer*, Vol. XXVIII., Schweizerbart'sche Verlagsbuchhandlung, Stuttgart, 1983; and B. Rothe, et al., "The Phylogenetic Position of the Budding Bacteria *Blastobacter aggregatus* and *Gemmobacter aquatilis*" in *Archives of Microbiology*, 147:92-99, 1987.

^bMany are mixotrophs.

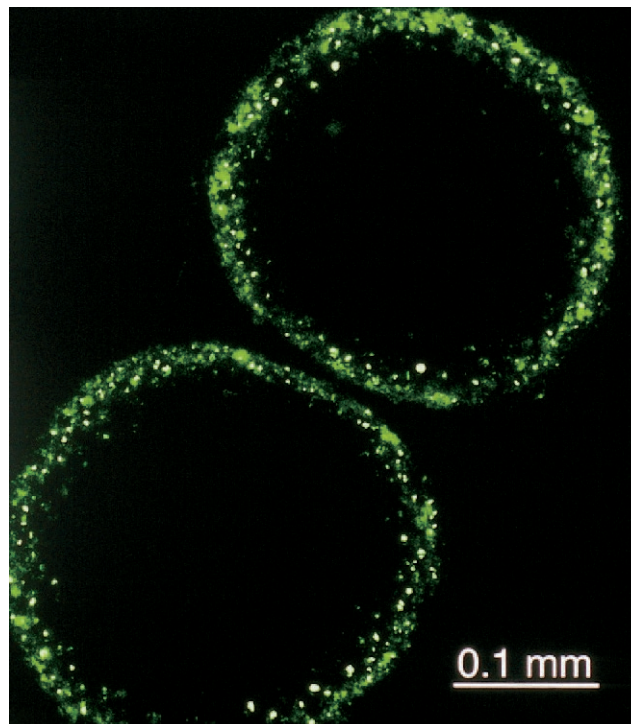


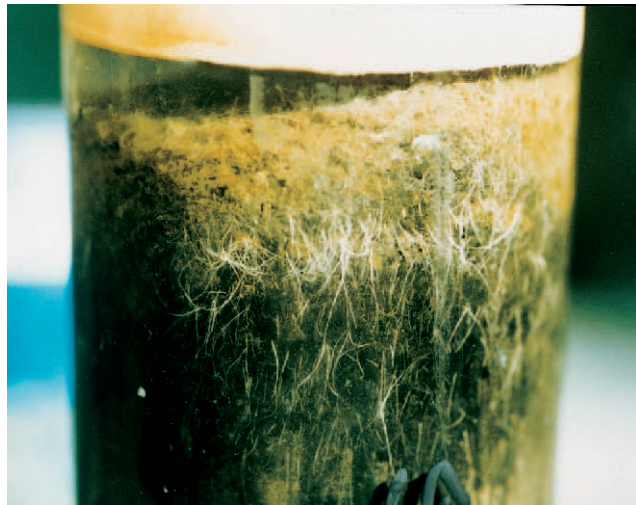
Figure 29.5 *Thiomargarita namibiensis*, the World's Known Largest Bacterium. This procaryote, usually 100 to 300 μm in diameter as shown here, occasionally reaches a size of 0.75 mm (larger than a period on this page), 100 times the size of a common bacterium. This unique bacterium uses sulfide from bottom sediments as an energy source and nitrate, which is found in the overlying waters, as an electron acceptor.

One of the most exciting findings in the last few years is the presence of extremely high levels of **ultramicrobacteria** or **nanobacteria** in aquatic environments, particularly the genus *Sphingomonas*, which easily can pass through a 0.2 μm membrane filter. These cells, which have a volume of less than 0.08 μm^3 , have been found to be the dominant bacteria in marine systems (and in soils, as will be discussed in the next chapter); they may approach 10^{12} to 10^{13} cells per gram or milliliter of material. They can make up the dominant microbial biovolume in many environments, as has been shown in the northern Pacific Ocean region. These ultramicrobacteria are so small that they are not grazed as efficiently by heterotrophic nanoflagellates, giving them a unique survival advantage.

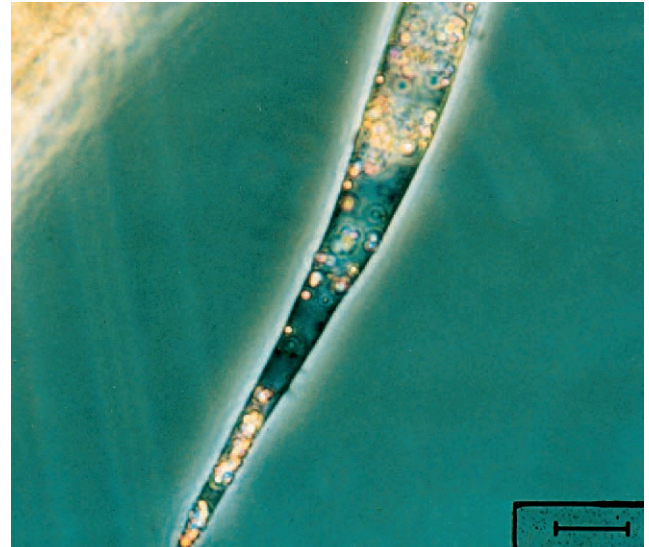
Another unusual marine microbe, found off the coast of Namibia on the west coast of Africa, is *Thiomargarita namibiensis* (figure 29.5; see also Box 3.1), which means the "sulfur pearl of Namibia." This microorganism is considered to be the world's largest bacterium! The individual cells are usually 100 to 300 μm in diameter (750 μm cells occasionally occur) and sulfide and ni-

trate are used as the reductant and oxidant, respectively. In this case the nitrate, from the overlying seawater, penetrates the anaerobic sulfide-containing muds only during storms. When this short-term mixing occurs, the *Thiomargarita* takes up and stores the nitrate in a huge internal vacuole, which can take up 98% of the organism's volume. The vacuolar nitrate can approach a concentration of 800 mM. The elemental sulfur granules appear near the cell edge in a thin layer of cytoplasm. Between the storms, the organism lives using the stored nitrate as an oxidant. These unique bacteria are important in sulfur and nitrogen cycling in these environments.

A critical adaptation of microorganisms in aquatic systems is the ability to link and use resources that are in separate locations, or that are available at the same location only for short intervals such as during storms. One of the most interesting bacteria linking widely separated resources is *Thioploca* spp., which lives in bundles surrounded by a common sheath (figure 29.6). These microbes are found in upwelling areas along the coast of Chile, where oxygen-poor but nitrate-rich waters are in contact with sul-



(a)



(b)

Figure 29.6 *Thioploca* the “Spaghetti Bacterium.” *Thioploca* (sulfur braid) is an unusual microorganism that links separated resources of sulfide from the mud and nitrate from the overlying water. (a) Bundles that join the aerobic surface and the lower anaerobic mud. (b) An individual *Thioploca*, showing the elemental sulfur globules and tapered ends. Bar = 40 μm .

fide-rich bottom muds. The individual cells are 15 to 40 μm in diameter and many centimeters long, making them one of the largest bacteria known. They form filamentous sheathed structures, and the individual cells can glide 5 to 15 cm deep into the sulfide-rich sediments. These unique microorganisms are found in vast fields off the coast of Chile and are “the largest communities of visible bacteria in the world.”

Other microorganisms take advantage of surfaces in aquatic systems. These include sessile microorganisms of the genera *Sphaerotilus* and *Leucothrix* and the prosthecate and budding bacteria of the genera *Caulobacter* and *Hyphomicrobium*. There are also a wide range of aerobic gliding bacteria such as the genera *Flexithrix* and *Flexibacter*, which move over surfaces where organic matter is adsorbed. These organisms are characterized by their exploitation of surfaces and nutrient gradients. They are obligate aerobes, although sometimes they can carry out denitrification, as occurs in the genus *Hyphomicrobium*. In addition, bacteria may be primarily colonizers of submerged surfaces, allowing subsequent development of a more complex biofilm. Budding bacterium (pp. 490–92); Sheathed bacteria (pp. 496–97); Gliding bacteria (pp. 482–83)

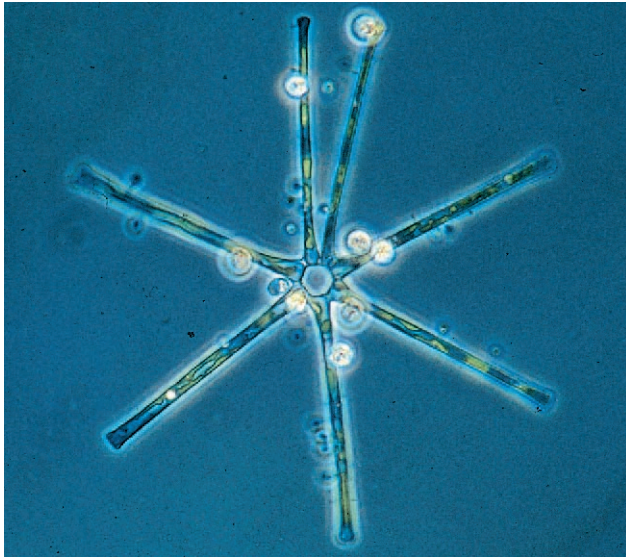
Microscopic fungi, which usually are thought to be terrestrial organisms living in soils and on fruits and other foods, also grow in freshwater and marine environments. Zoospore fungi that are adapted to an aquatic existence include both the oomycetes, which have motile asexual reproductive spores with two flagella, and the chytrids, which have motile asexual reproductive spores

with a single whiplash flagellum (see chapter 25). Most chytrids are important because of their role in decomposing dead organic matter, but they also cause potato wart disease and parasitize invertebrates, particularly nematodes and mosquitoes. In addition, many chytrids attack algae (figure 29.7).

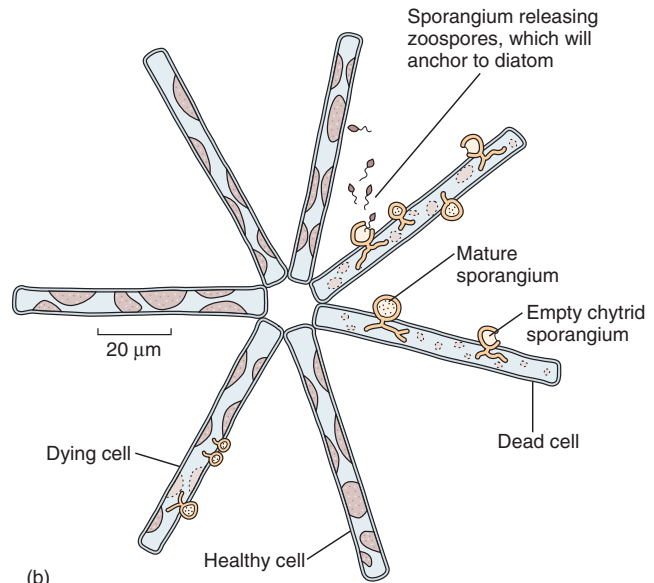
Recently chytrids have been reported to live in the skin cells of amphibians, and these have been associated with die-offs of frogs and toads in many parts of the world, including the United States, Australia, and Central America. The chytrid *Batrachochytrium dendrobatidis* is one of the major zoospore fungi that has been associated with amphibian mortalities.

Another important group includes filamentous fungi that can sporulate under water. These hypomycetes include the **Ingoldian fungi**, named after C. T. Ingold. In 1942 Ingold discovered fungi that produce unique tetradiate forms (figure 29.8a). The ecology of these aquatic fungi is very interesting (figure 29.8b). The tetradiate conidium forms on a vegetative mycelium, which grows inside decomposing leaves. Released tetradiate conidia are transported by the water and often are present in surface foam. When they contact leaves, the conidia attach and establish new centers of growth. These uniquely adapted fungi contribute significantly to the processing of organic matter, especially leaves. Often aquatic insects will feed only on leaves that contain fungi. Without fungal processing, the insects are not interested!

One of the most interesting recent discoveries regarding the marine environment is the presence of large numbers of archaea. By use of rRNA sequence comparisons (see chapter 19), it has

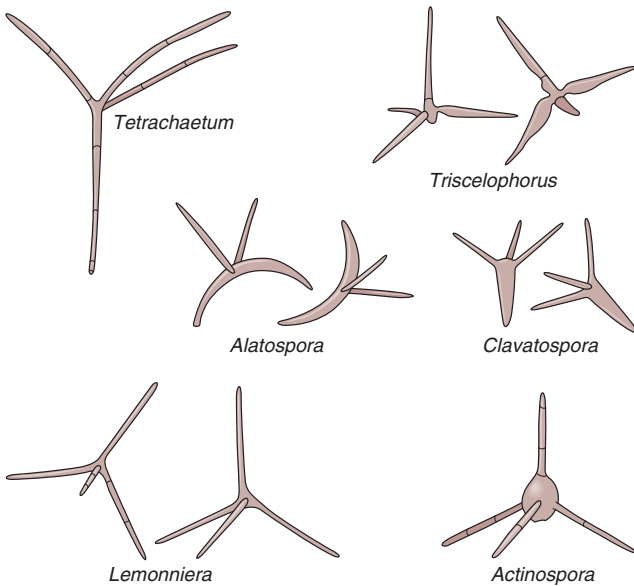


(a)

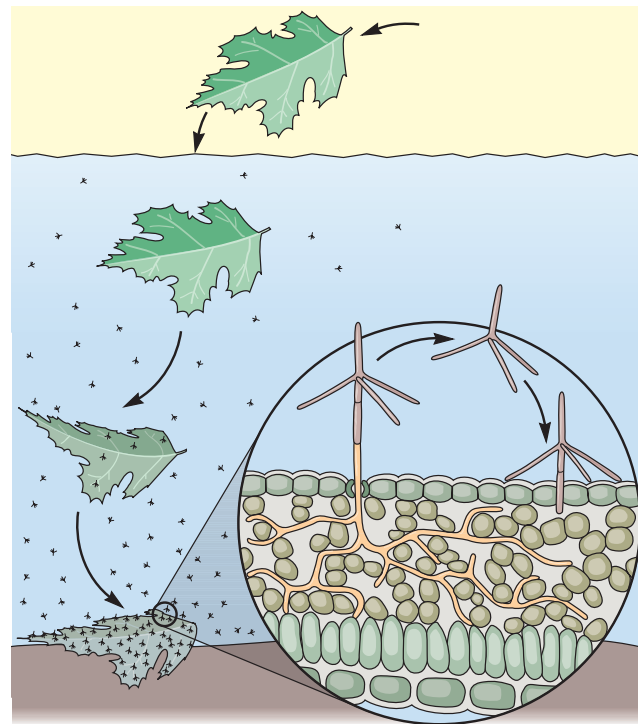


(b)

Figure 29.7 Chytrids and Aquatic Environments. Chytrids play important roles in aquatic environments. (a) The infection of the diatom by the chytrid *Rhizophydium* is shown in the photograph, and (b) in the illustration showing details of the parasitism process.



(a)



(b)

Figure 29.8 Ingoldian Fungi. These fungi, named after C. T. Ingold, are capable of sporulation under water. These aquatic fungi play important roles in the processing of complex organic matter, such as leaves, which fall into streams and lakes. These microbes include types with tetra- and triradiate conidia (a). The tetra- and triradiate conidia are produced as aerial structures (b) from the mycelium, which is growing inside the decomposing leaf. The new tetra- and triradiate conidium then can be released and attaches to a leaf surface, repeating the process and accelerating leaf decomposition.

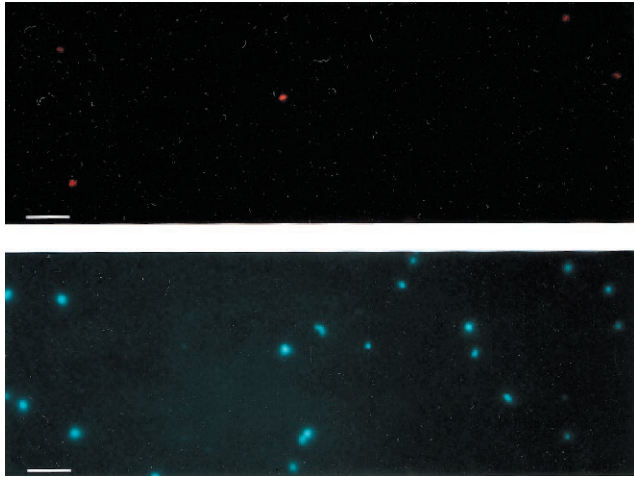


Figure 29.9 Detection of Archaea in the Ocean with Probes.

Fluorescent-labeled probes and selective barrier filters can be used to detect archaea (top plate) as a part of a mixed population of procaryotes (bottom plate) in deep marine water samples using microscopy. Bar = 5 μm .

been determined that approximately 1/3 of the oceanic picoplankton (cells smaller than 2 μm) are archaea, particularly crenarchaeans, organisms normally associated with hostile environments (hot springs, deep marine “black smoker” areas, saline and thermoacidophilic regions).

Planktonic archaea and bacteria are present in both freshwater and marine environments, and deep in the ocean. Using fluorochrome labeled probes in a fluorescent in situ hybridization (FISH) technique, it is possible to detect these different microbial groups in the same sample by the use of different exciting wavelengths (**figure 29.9**). Bacterial concentrations are higher in the surface zone, but below 100 meters archaea form a greater portion of the total population (**figure 29.10**). More than 90% of the cells react to the stains and can be detected by the use of different probes and wavelengths.

Aquatic environments also harbor large populations of viruses. These are present at 10-fold higher concentrations than the bacteria, and most are bacteriophages. These **virio**plankton are an important part of the aquatic microbial community. They may influence the functioning of the microbial loop, be involved in horizontal gene transfer between procaryotes, and control microbial community diversity.

Microorganisms are constantly mixing and being added to waters. Aquatic microorganisms are released to the atmosphere and returned to other aquatic locations by air movement, a part of aerobiology. In rivers, microorganisms can be moved thousands of miles from high mountain areas to the ocean; in lakes, flushing and turnover can occur; and in marine environments, movement and cycling of waters can occur over centuries. Microorganisms associated with detritus, animals, and fish carcasses also move with their substratum. Occasionally, carcasses of large animals

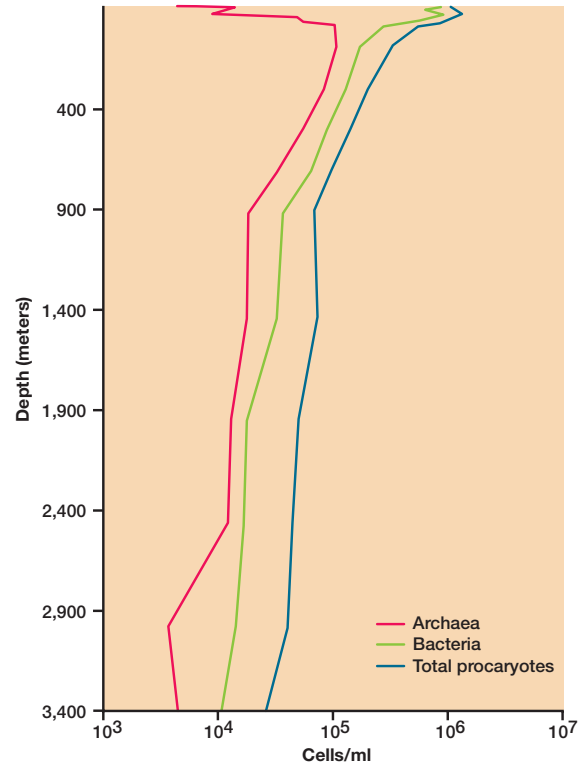
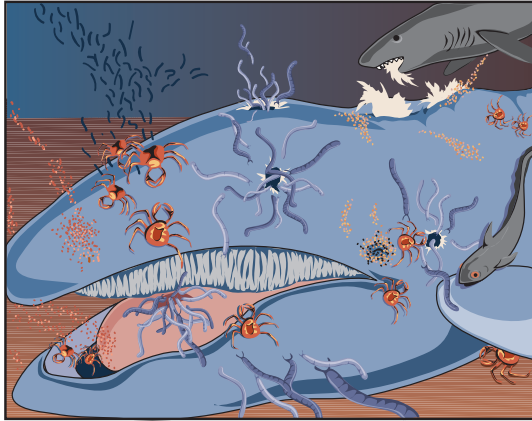


Figure 29.10 Archaea Are Plentiful in Ocean Depths. The distribution of group I archaea and bacteria, together with an estimate of total procaryotes, over a depth of 3,400 meters are shown at a Pacific Ocean location. These results indicate that archaea make up a significant part of the observable picoplankton below the surface zone.

such as whales will drop to the ocean floor over thousands of meters of water depth. The carcass of a dead whale lands on the ocean floor, creating new opportunities for scavengers and microorganisms (**figure 29.11**). The mobile scavenger stage lasts 0-6 months (**figure 29.11a**); the sulfophilic state lasts 1 to 2 years, with sulfides released from bones allowing chemoautotrophic microbial growth (**figure 29.11b**); and finally after 2 to 3 years, the enrichment opportunist stage occurs (**figure 29.11c**). These transfers result in the continuous inoculation of oceans, especially, with microorganisms from other regions of the Earth.

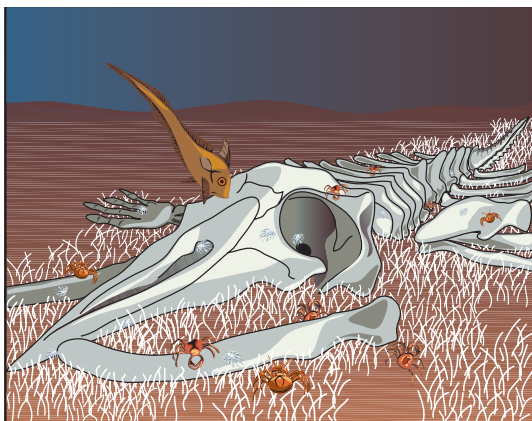
Dust and sediment from distant terrestrial areas also continually fall on all waters and ice-covered regions, due to wind and storms, another important part of aerobiological processes. Through these atmospheric transport processes, microorganisms are constantly being mixed across all regions of the Earth. In this regard, recent studies indicate that microorganisms cultured from sediments at the bottom of the deep ocean are similar to those from surface soils. Another example is airborne dust from the sub-Saharan region of Africa, which has been linked to the massive die-off of corals in the Caribbean due to fungi and algae contained in soils that have drifted around the world in the atmosphere.



(a)



(b)



(c)

Figure 29.11 The Fate of a Whale Carcass in the Ocean Deep. Materials can move great distances in aquatic environments and transfer nutrients and microorganisms to areas where they do not normally occur, creating oases with altered microbial communities. (a) The mobile scavenger phase (0 to 6 months). (b) The sulfophilic state, lasting 1 to 2 years, where chemoautotrophic microorganisms develop. (c) The enrichment opportunist stage. See text for details. Adapted by permission from paintings by Michael Rothman.

1. Discuss the unusual structure and function of *Thioploca*. What environment is this organism found in?
2. What does *Thiomargarita* mean? Where would one go to find this interesting microorganism?
3. Give the genera of sessile and prosthecate bacteria that are important in aquatic environments.
4. What are chytrids? Why might they be important in aquatic environments?
5. Describe Ingoldian fungi. How do they function in aquatic environments, and what is unique about their shape? What does the term tetrastrate mean?
6. What levels of archaea are found in aquatic environments? How have these been detected?
7. How are microorganisms moved within and between various water bodies?

29.3 Marine Environments

In terms of sheer volume, the marine environment represents a major portion of the biosphere and contains 97% of the Earth's water. Much of this is in the deep sea at a depth greater than 1,000 meters, representing 75% of the ocean's volume. The ocean has been called a "high-pressure refrigerator," with most of the volume below 100 meters at a constant 3°C temperature. The ocean, at its greatest depth, is slightly more than 11,000 meters deep or equivalent to almost 29 Empire State Buildings (each 1,250 feet or 381 meters in height) stacked on top of one another! The pressure in the marine environment increases approximately 1 atm/10 meters in depth, and pressures are in the vicinity of 1,000 atm at the greatest ocean depths (**figure 29.12**).

A series of pressure relationships is observed among the bacteria growing in this vertically differentiated system. Some bacteria are barotolerant and grow between 0 and 400 atm, but best at normal atmospheric pressure. Many other bacteria are **barophiles** [Greek, *baro*, weight and *philein*, to love] and prefer higher pressures. Moderate barophiles grow optimally at 400 atm, but still grow at 1 atm; extreme barophiles grow only at higher pressures. Pressure differences influence many biological processes including cell division, flagellar assembly, DNA replication, membrane transport, and protein synthesis. Porins, outer membrane proteins that form channels for diffusion of materials into the periplasm (*see figure 3.23*), also function most effectively at specific pressures. [Pressure and microbial growth \(p. 129\)](#)

Most nutrient cycling in oceans occurs in the top 300 meters where light penetrates. Light allows phytoplankton to grow and fall as a "marine snow" to the seabed. This "trip" can take a month or longer. Most of the organic matter that falls below the 300-meter zone is decomposed, and only 1% of photosynthetically derived material reaches the deep-sea floor unaltered. Because low inputs of organic matter occur in the deep sea, the ability of microorganisms to grow under oligotrophic conditions becomes important (*see section 6.5*).

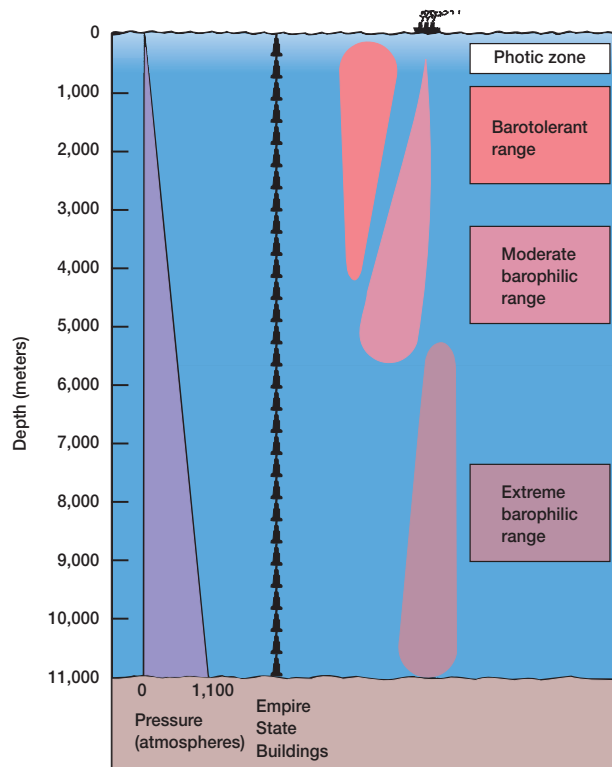


Figure 29.12 The Deep Marine Environment. Microorganisms with specialized pressure relationships are found at various depths. The relative activity and depth relationships of barotolerant, moderate barophilic, and extreme barophilic microorganisms are noted schematically. The pressure can approach 1,100 atmospheres at the deepest ocean depths. The depths are given in meters and “stacked” Empire State Buildings for perspective. Light penetrates only into a relatively shallow surface layer, creating the photic zone.

The carbon cycle within the ocean environment is only poorly understood (figure 29.13), but clearly microorganisms can profoundly influence the cycle. It is estimated that the large amount of dissolved organic carbon (DOC) in the ocean has a mean age of greater than 1,000 years. Organic matter in deep ocean waters has a similarly long residence time. Besides DOC, massive deposits of methane hydrate occur in ocean sediments. Under the low-temperature, high-pressure conditions found at the ocean floor below 500 meters, methane accumulates in latticelike cages of crystalline water. There may be up to 10,000 billion metric tons of carbon present as methane hydrate worldwide, 80,000 times the world’s current known natural gas reserves.

The nitrogen and sulfur cycles also function at an “ocean scale” and have potentially significant effects on global-level processes, which have not been appreciated until recently. Large volumes of the ocean’s water have lower oxygen levels, leading to denitrification and a decrease in the nitrate-phosphate ratio in the water. As a consequence, nitrogen fixation may be favored

and increase the nitrogen level in the water. It appears that nitrogen, and not phosphorus, often can limit biological activity in many marine environments. The ocean sulfur cycle also has widespread effects on global processes. Dimethylsulfide (DMS), an algal osmolite, can be released to the atmosphere and comprises 90% of the volatile sulfur compounds in the sulfur cycle. When DMS is oxidized, its end products can influence the acidity of the atmosphere, as well as the Earth’s temperature and cloud formation.

Additional cold-temperature microorganisms include those involved in the production and use of methane. As already noted, methane hydrates are found at low temperatures and high pressures in many marine regions of the world. The procaryotes that consume these methane hydrates serve as a food source for ice worms, *Hesiocaeca methanicola*. Archaea, as a part of complex microbial communities, appear to metabolize methane deposits at low hydrogen levels under sulfate reducing conditions. The process is called **reverse methanogenesis**.

Much of the marine environment is covered by sea ice that may comprise up to 7% of the world’s surface at the winter maxima at the north and south poles! Microorganisms actually grow and reproduce at the interface between the ice and the seawater. **Figure 29.14** shows an ice core taken from the lower zone of an ice sheet at the seawater-ice interface; a band of microorganisms is clearly visible in the ice profile. Pockets of salt brine also form in this region, creating additional environments with varied temperatures, salinities, nutrient concentrations, and light levels for these cold-adapted microorganisms. The microbes that have been recovered from these ice cores have been given intriguing generic names: these include *Polaromonas*, *Marinobacter*, *Psychroflexus*, *Iceobacter*, *Polibacter*, and *Psychromonas antarcticus*.

Increased human populations and the urban development occurring in coastal areas around the world is taxing the seemingly inexhaustible ability of oceans to absorb and process pollutants. Coastal areas that have limited mixing with ocean waters (e.g., the Baltic Sea, Long Island Sound, Chesapeake Bay, the Mediterranean) are showing signs of nutrient enrichment and microbial pollution. One example is shellfish contamination by runoff waters from urbanized coastal areas. Only a few years ago shellfish could be harvested without delay after major rainfalls; now, one week or longer is needed to allow dieback of polluting microorganisms. This economically impacts individuals who depend on harvesting of shellfish for a livelihood.

In the Gulf of Mexico, at the Mississippi River delta, releases of nutrients from states that drain into this river have stimulated microbial growth and oxygen depletion. This has created a “dead zone,” a hypoxic or anoxic region, that is larger than the state of New Jersey. The lowered oxygen levels occurring in this rich shellfish area have damaged the economy of these aquaculture-dependent regions.

Another problem that relates to ocean waters and water mixing in coastal areas is the occurrence of red tides (see Box 26.2). This has major economic effects when shellfish cannot be harvested or consumed. Recently the occurrence of algal blooms and red tides in the Pacific Ocean off the central California coast has resulted in the die-off of dolphins. The major algal genus responsible for the

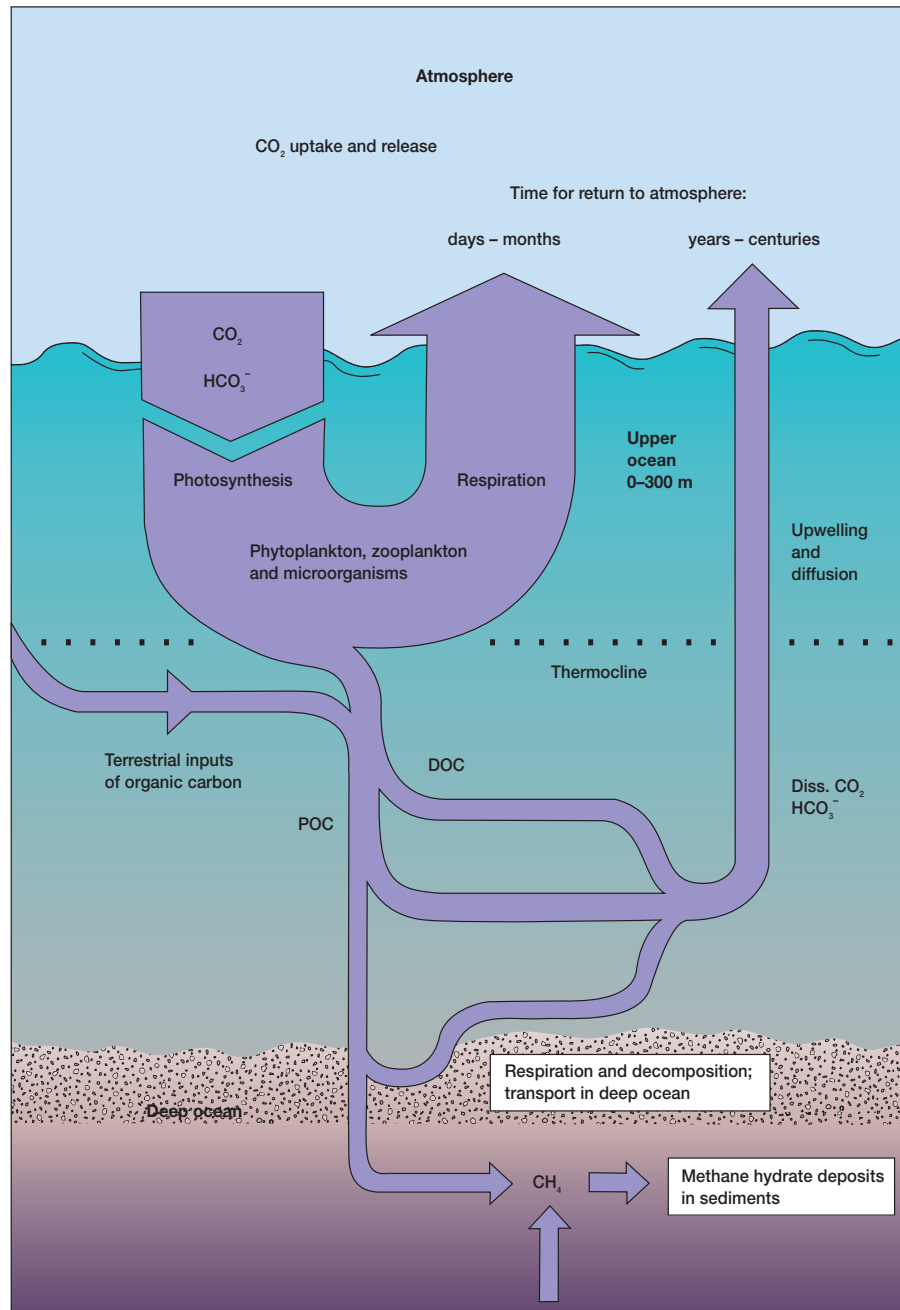


Figure 29.13 Carbon Cycling in the Ocean Environment. Microorganisms in the oceans can influence global carbon cycling and ocean-atmosphere interactions. Most carbon processing occurs in the surface water zone, with particulate organic carbon (POC), dissolved organic carbon (DOC), and methane hydrate (in sediments) being major carbon pools. The ocean also contains bicarbonate and dissolved CO₂ (diss. CO₂) that come from the atmosphere and the degradation of organic carbon. Methane hydrate allows microorganisms and associated animals such as ice worms to develop.

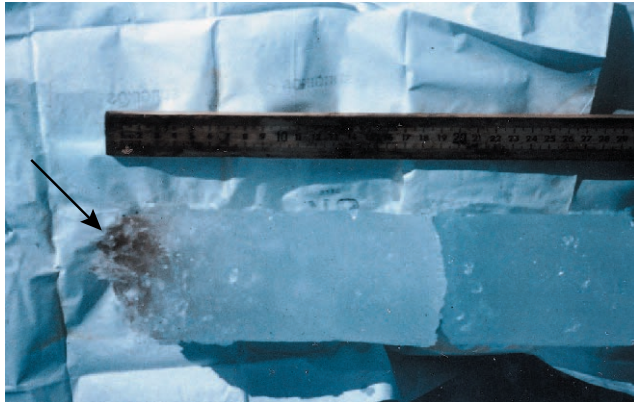


Figure 29.14 Sea ice as a Habitat for Microorganisms. Sea ice allows the development of complex microbial communities. The bottom of a sea ice core from the Antarctic, which is in contact with the underlying seawater, is shown. The dark band (see arrow) is the sea ice microbial community.

widespread loss of these aquatic animals is *Pseudo-nitzschia*. The key link is anchovies. This fish, consumed by the dolphins, feeds on algae that contain a high concentration of a neurotoxin, domoic acid; this toxin has been detected in the sick and dead dolphins.

Algae may cause problems other than red tides. An excellent example is *Pfiesteria piscicida*, a dinoflagellate (see pp. 579–80) that has produced major fish kills from Maryland southward along the U.S. Atlantic coast. It even can cause dizziness and loss of memory in people exposed to its toxins. *Pfiesteria* has an exceptionally complex life cycle that probably involves at least 24 stages (figure 29.15)! This primitive protistan alga is a fish predator. Its flagellated vegetative cell stage can detect fish compounds in the water and then literally attack the approaching fish. *Pfiesteria* produces at least two powerful toxins: one that stuns the fish and another that causes the characteristic lesions (figure 29.16). This dinoflagellate apparently once used other algae as its primary food source, but now has become a killer of fish, eels, crabs, and other animals. Increased nutrient levels are suspected as the cause of its increasing prevalence.

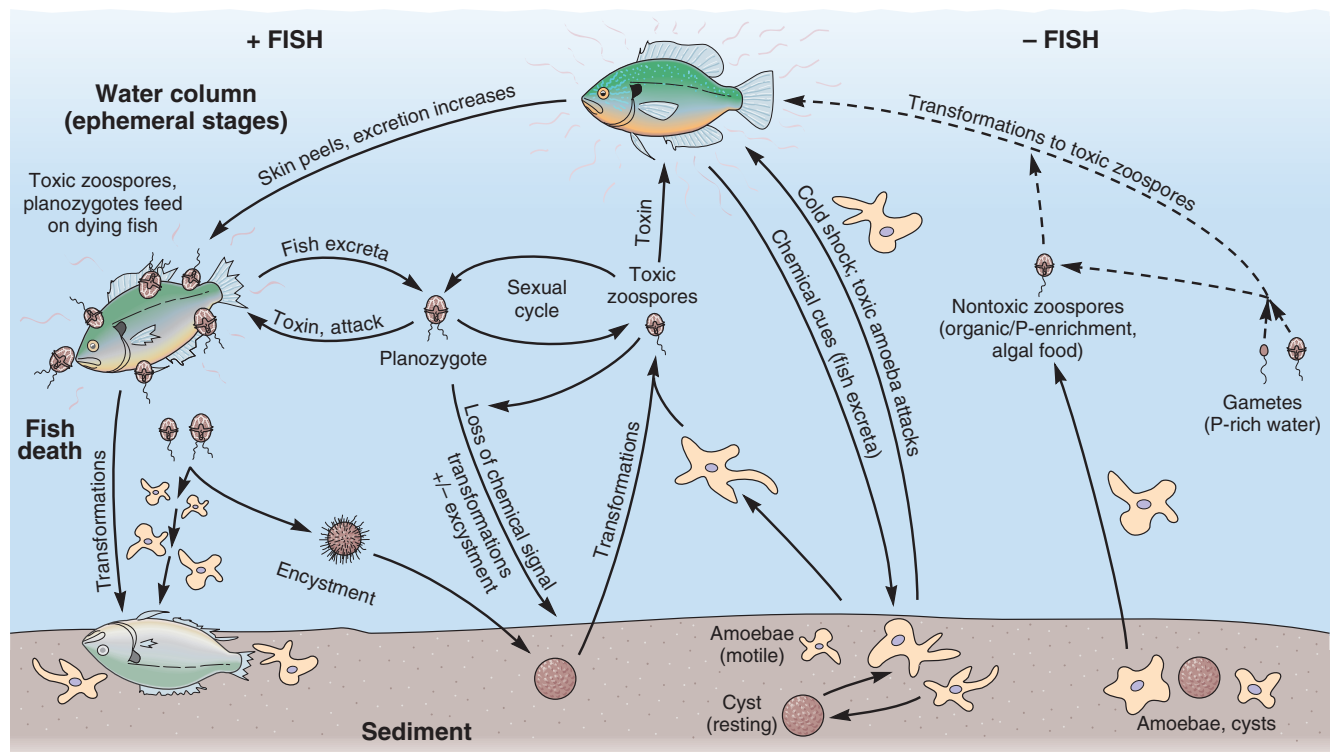


Figure 29.15 Basic Life Cycle of *Pfiesteria piscicida*. The cycle can occur with (+) and without (–) the presence of a fish in the overlying waters.



Figure 29.16 *Pfiesteria* Lesions. Lesions on menhaden resulting from parasitism by the dinoflagellate *Pfiesteria piscicida*. Note that the worker is using thick gloves—humans also are susceptible to these toxins, which can cause dizziness, loss of memory, and impaired motor function.

Global-level movements of air also affect marine microorganisms. The atmospheric movement of soils and industrial activities influence phytoplankton growth and their Redfield ratio (p. 638). In the northern Pacific, the water is iron-limited, and iron is being added to these waters by desertification and dust storms in central Asia, thus increasing primary production. In contrast, the northern Atlantic Ocean is nitrogen-limited, and a transition from N- to P-limitation is occurring with increased depositions of atmospheric nitrogen from human activities. The N:P ratio in the deep North Atlantic is increasing and altering the phytoplankton Redfield ratio.

1. What portion of the water in the world is marine?
2. Why is sea ice an important habitat for microorganisms? Where are the microorganisms mostly located?
3. What microbially produced volatile sulfur compound can influence the weather?
4. Where are methane hydrates found, and what is reverse methanogenesis?
5. What is hypoxia or anoxia? What are possible causes?
6. Microbiological quality of waters in marine coastal areas is important. What are some major concerns?
7. What link has been found recently between algae, anchovies, and dolphins?
8. What is unique about *Pfiesteria*?
9. How can atmospheric processes influence nutrient limitation in marine waters?

29.4 Freshwater Environments

Most fresh water that is not locked up in ice sheets, glaciers, or groundwaters is found in lakes and rivers. These provide microbial environments that are different from the larger oceanic systems in many important ways. For example, in lakes, mixing and

water exchange can be limited. This creates vertical gradients over much shorter distances. Changes in rivers occur over distance and/or time as water flows through river channels.

Lakes

Lakes vary in nutrient status. Some are **oligotrophic** or nutrient-poor (**figure 29.17a**), others are **eutrophic** or nutrient-rich (**figure 29.17b**). Nutrient-poor lakes remain aerobic throughout the year, and seasonal temperature shifts do not result in distinct oxygen stratification. In contrast, nutrient-rich lakes usually have bottom sediments that contain organic matter. In thermally stratified lakes the **epilimnion** (warm, upper layer) is aerobic, while the **hypolimnion** (deeper, colder, bottom layer) often is anaerobic (particularly if the lake is nutrient-rich). The epilimnion and hypolimnion are separated by a zone of rapid temperature decrease called the **thermocline**, and there is little mixing of water between the two layers. In the spring and fall, the aerobic surface water and the anaerobic subsurface water will turn over as the result of differences in temperature and specific gravity. After such mixing occurs, motile bacteria and algae migrate within the water column to again find their most suitable environment.

When sufficiently large amounts of nutrients are added to water, **eutrophication** (nutrient enrichment) takes place and stimulates the growth of plants, algae, and bacteria (*see figure 21.11*). Because nitrogen and phosphorus frequently limit microbial growth in freshwater habitats, the addition of nitrogen and phosphorus compounds has a particularly large impact on freshwater systems. Depending on the body of water and the rate of nutrient addition, eutrophication may either require many centuries or occur very rapidly.

If phosphorus is added to oligotrophic fresh water, cyanobacteria play the major role in nutrient accumulation, even in the absence of extra nitrogen. Several genera, notably *Anabaena*, *Nostoc*, and *Cylindrospermum*, can fix nitrogen under aerobic conditions (*see section 21.3*). The genus *Oscillatoria*, using hydrogen sulfide as an electron donor for photosynthesis, can fix nitrogen under anaerobic conditions. Even with both nitrogen and phosphorus present, cyanobacteria still can compete with algae. Cyanobacteria function more efficiently with higher pH conditions (8.5 to 9.5) and higher temperatures (30 to 35°C). Eucaryotic algae, in comparison, generally prefer a more neutral pH and have lower optimum temperatures. By using CO₂ at rapid rates, cyanobacteria also increase the pH, making the environment less suitable for eucaryotic algae. [Algae \(chapter 26\); Cyanobacteria \(pp. 471–76\)](#)

Cyanobacteria have additional competitive advantages. Many produce hydroxamates, which bind iron, making this important trace nutrient less available for the eucaryotic algae. Cyanobacteria also often resist predation because of their production of toxins. In addition, some synthesize odor-producing compounds that affect the quality of drinking water.

Both cyanobacteria and algae can contribute to massive blooms in strongly eutrophied lakes (*see figure 21.11*). This problem may continue for many years, until the nutrients are eventually lost from the lake by normal water passage or by precipitation

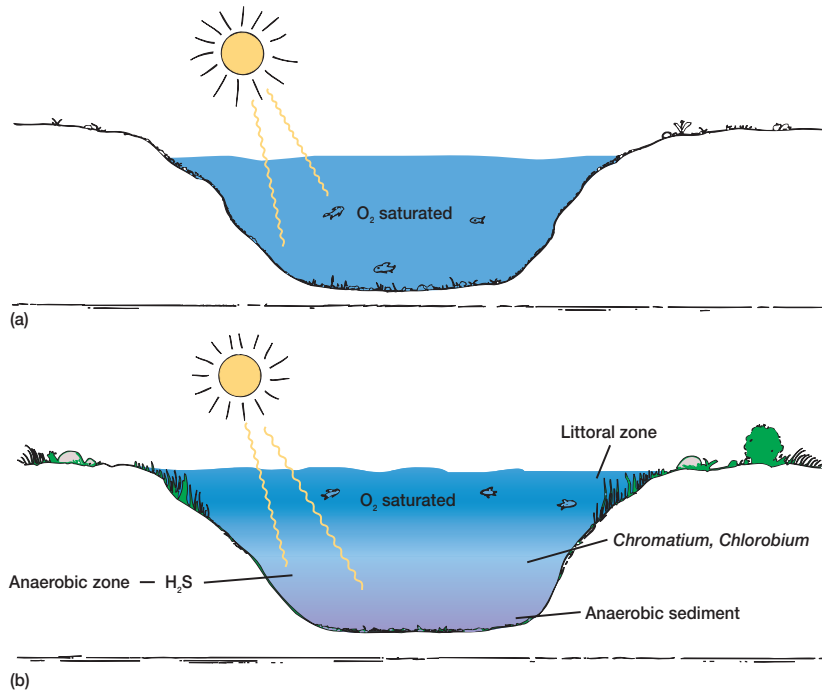


Figure 29.17 Oligotrophic and Eutrophic Lakes. Lakes can have different levels of nutrients, ranging from low nutrient to extremely high nutrient systems. The comparison of (a) an oligotrophic (nutrient-poor) lake, which is oxygen saturated and has a low microbial population, with (b) a eutrophic (nutrient-rich) lake. The eutrophic lake has a bottom sediment layer and can have an anaerobic hypolimnion. Sulfur photosynthetic microorganisms can develop in this anaerobic region.

of nutrients in bottom sediments. Lake management can improve the situation by removing or sealing bottom sediments or adding coagulating agents to speed sedimentation.

1. What terms can be used to describe the different parts of a lake?
2. Why are cyanobacteria so important in waters that have been polluted by phosphorus additions?
3. What are some important effects of eutrophication on lakes?

Streams and Rivers

Streams and rivers present a different situation from lakes in that there is sufficient horizontal water movement to minimize vertical stratification; in addition, most of the functional microbial biomass is attached to surfaces. Only in the largest rivers will a greater relative portion of the microbial biomass be suspended in the water. Depending on the size of the stream or river, the source of nutrients may vary. The source may be in-stream production based on photosynthetic microorganisms (figure 29.18a). Nutrients also may come from outside the stream, including runoff sediment from riparian areas (the edge of a river), or leaves and other organic matter falling directly into the water (figure 29.18b). Chemoorganotrophic microorganisms metabolize the available organic material and provide an energy base for the ecosystem. Under most conditions the amounts of organic matter added to streams and rivers will not exceed the system's oxidative capacity, and productive, aesthetically pleasing streams and rivers will be maintained.

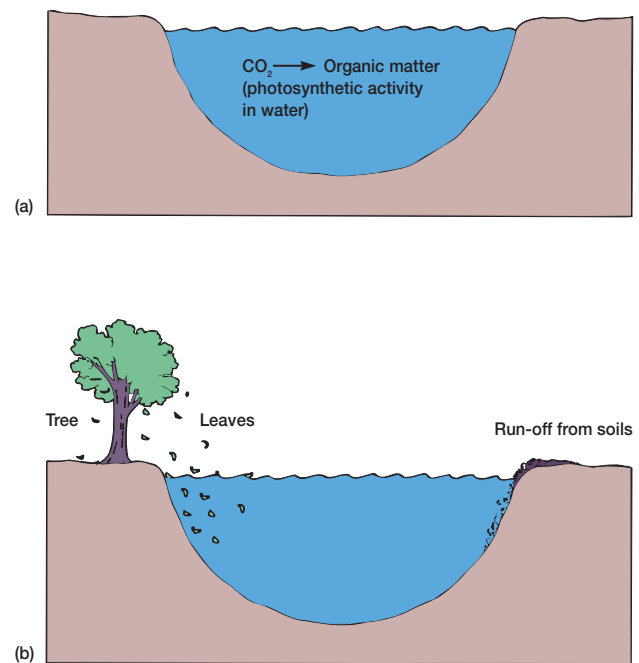
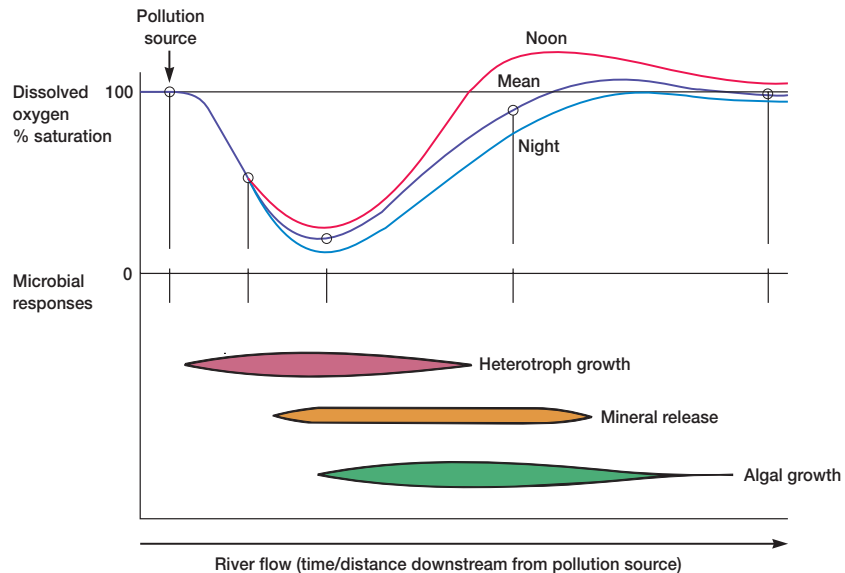


Figure 29.18 Organic Matter Sources for Lakes and Rivers. Organic matter used by microorganisms in lakes and rivers can be synthesized in the water, or can be added to the water from outside. (a) Within-stream sources of organic matter, primarily photosynthesis, and (b) sources of organic matter which enter streams and rivers from outside the water. Stream cross sections are shown.

Figure 29.19 The Dissolved Oxygen Sag Curve. Microorganisms and their activities can create gradients over distance and time when nutrients are added to rivers. An excellent example is the dissolved oxygen sag curve, caused when organic wastes are added to a clean river system. During the later stages of self-purification, the phototrophic community will again become dominant, resulting in diurnal changes in river oxygen levels.



The capacity of streams and rivers to process such added organic matter is, however, limited. If too much organic matter is added, the water may become anaerobic. This is especially the case with urban and agricultural areas located adjacent to streams and rivers. The release of inadequately treated municipal wastes and other materials from a specific location along a river or stream represents a **point source of pollution**. Such point source additions of organic matter can produce distinct and predictable changes in the microbial community and available oxygen, creating an oxygen sag curve (figure 29.19). Runoff from fields and feedlots, which causes algal blooms in eutrophic water bodies, is an example of a **nonpoint source of pollution**.

When the amount of organic matter added is not excessive, the algae will grow using the minerals released from the organic matter. This leads to the production of O_2 during the daylight hours, and respiration will occur at night farther down the river, resulting in **diurnal oxygen shifts**. Eventually the O_2 level approaches saturation, completing the self-purification process.

Along with the stresses of added nutrients, removal of silicon from rivers by the construction of dams and trapping of sediments causes major ecological disturbances. For example, construction of the dam at the “iron gates” on the Danube (600 miles above the Black Sea) has led to a decrease in silicon to 1/60th of the previous concentration. This decreased silicon availability then inhibits the growth of diatoms (see pp. 577–78) because the ratio of silicon to nitrate has been altered (silicon is required for diatom frustule formation). With this shift in resources, Black Sea diatoms are not able to grow and immobilize nutrients. The result has been a 600-fold increase in nitrate levels and a massive development of toxic algae. Thus the delicate balance of rivers can be altered in unexpected ways by dams (there are over 36,000 dams in the world, and more being built, including in China), leading to effects on aquatic microbiological processes and whole ecosystems.

With the damaging effects of dams now being recognized, there is a growing trend to attempt to breach these structures to

restore normal water and sediment movements, and to allow fish migration into upper regions of rivers where they often have been excluded for decades.

Microorganisms in Freshwater Ice

Fresh waters include glaciers, as well as vast ice sheets in the Arctic and Antarctic polar regions. These regions, although far away from most people, are important as habitats for microorganisms. Of particular interest are deep frozen lakes in the Antarctic region. An excellent example is the McMurdo Dry Valley Lakes, which have 3 to 6 meter thick ice layers with windblown sediments occupying various levels in the ice profile. In the summer, melting of the ice in these sediment-containing zones provides an opportunity for microbial growth. There is interest in studying ice-inhabiting microorganisms from the north and south poles. Are they isolated and different populations? Do some of them show biogeographic distribution?

One of the most interesting deep-frozen habitats lies above Lake Vostok in the Antarctic, where cores have been taken to a depth of 3,600 meters (11,800 feet) in ice that has been frozen for over 420,000 years. The actual Lake Vostok lies below another 120 meters of ice. The lake itself has a size similar to that of Lake Ontario, but it is deeper, being 500 meters in depth. Scientists are actively searching for microorganisms in these unique older freshwater frozen environments. It is hoped that it will be possible to link microbial communities and the time of their deposition.

1. What is an oxygen sag curve, and what changes in a river lead to these effects?
2. What are point and nonpoint pollution? Give examples.
3. Why might dams influence microorganisms and microbial processes in rivers?
4. What is unique about Lake Vostok in the Antarctic?

Table 29.3 Water-Based Microbial Pathogens That Can Be Maintained in the Environment Independent of Humans

Organism	Reservoir	Comments
Bacteria		
<i>Aeromonas hydrophila</i>	Free-living	Sometimes associated with gastroenteritis, cellulitis, and other diseases
<i>Campylobacter</i>	Bird and animal reservoirs	Major cause of diarrhea; common in processed poultry; a microaerophile
<i>Helicobacter pylori</i>	Free-living, not known	Can cause type B gastritis, peptic ulcers, gastric adenocarcinomas
<i>Legionella pneumophila</i>	Free-living and associated with protozoa	Found in cooling towers, evaporators, condensers, showers, and other water sources
<i>Leptospira</i>	Infected animals	Hemorrhagic effects, jaundice
<i>Mycobacterium</i>	Infected animals and free-living	Complex recovery procedure required
<i>Pseudomonas aeruginosa</i>	Free-living	Swimmer's ear and related infections
<i>Salmonella enteritidis</i>	Animal intestinal tracts	Common in many waters
<i>Vibrio cholera</i>	Free-living	Found in many waters and estuaries
<i>Vibrio parahaemolyticus</i>	Free-living in coastal waters	Causes diarrhea in shellfish consumers
<i>Yersinia enterocolitica</i>	Frequent in animals and in the environment	Waterborne gastroenteritis
Protozoa		
<i>Acanthamoeba</i>	Sewage sludge disposal areas	Can cause granulomatous amebic encephalitis (GAE); keratitis, corneal ulcers
<i>Cryptosporidium</i>	Many species of domestic and wild animals	Causes acute enterocolitis; important with immunologically compromised individuals; cysts resistant to chemical disinfection; not antibiotic sensitive
<i>Cyclospora cayetanensis</i>	Waters—does not withstand drying; possibly other reservoirs	Causes long-lasting (43 days average) diarrheal illness; infection self-limiting in immunocompetent hosts; sensitive to prompt treatment with Bactrim
<i>Giardia lamblia</i>	Beavers, sheep, dogs, cats	Major cause of early spring diarrhea; important in cold mountain water
<i>Naegleria fowleri</i>	Warm water (hot tubs), swimming pools, lakes	Inhalation in nasal passages; central nervous system infection; causes primary amebic meningoencephalitis (PAM)

29.5 Waters and Disease Transmission

Since the beginning of recorded history water has been recognized as a potential carrier of disease. In order to protect his health, Alexander the Great (356–323 B.C.) had his personal drinking water carried in silver urns. Clearly the association between noble heavy metals such as silver and the prevention of waterborne diseases was early established through chance observation. The connection between a freshwater supply and the health of an urban population was recognized by the time of the Roman Empire (27 B.C.). However, much of the technology for the protection of the water supply was subsequently lost until the middle of the nineteenth century.

With increased use of waters by humans, especially as a receptacle for the disposal of human wastes, the effects of added organic matter and pathogens is of continuing concern for human health.

Waterborne Pathogens and Water Purification

Many important human pathogens are maintained in association with living organisms other than humans, including many wild animals and birds. Some of these bacterial and protozoan pathogens can survive in water and infect humans. As examples, *Vibrio vulnificus*, *V. parahaemolyticus*, and *Legionella* are of continuing concern. When waters are used for recreation or are a source of seafood that is consumed uncooked, the possibility for disease transmission

certainly exists. A waterborne infection may have serious consequences for immunologically compromised individuals.

Some major waterborne diseases are summarized in **table 29.3**. Waterborne bacterial and viral pathogens are extensively discussed in sections 38.4 and 39.4. The protozoan pathogens *Giardia*, *Cryptosporidium* are described in section 40.2; *Cyclospora* is discussed on page 653.

Another waterborne protozoan disease of increasing concern worldwide is primary amebic meningoencephalitis (PAM), an infection of the central nervous system caused by *Naegleria fowleri*. The disease usually occurs in children or young adults who have been swimming in lakes or pools or have been waterskiing. After nasal infection the protozoan reaches the brain and initiates an inflammatory response. The result usually is fatal. Warm water and heated industrial effluents promote the growth of this protozoan. Protozoan diseases (pp. 950–58); Food-borne and waterborne viral diseases (pp. 891–93); Food-borne and waterborne bacterial diseases (pp. 926–33)

Water purification is a critical link in controlling disease transmission in waters. As shown in **figure 29.20**, water purification can involve a variety of steps, depending on the type of impurities in the raw water source. For example, if the raw water contains large amounts of iron and manganese, which will often precipitate when water is exposed to air, it may be necessary to aerate the water and employ other methods to remove these ions early in the purification sequence. Usually municipal water supplies are purified by a process that consists of at least three or four

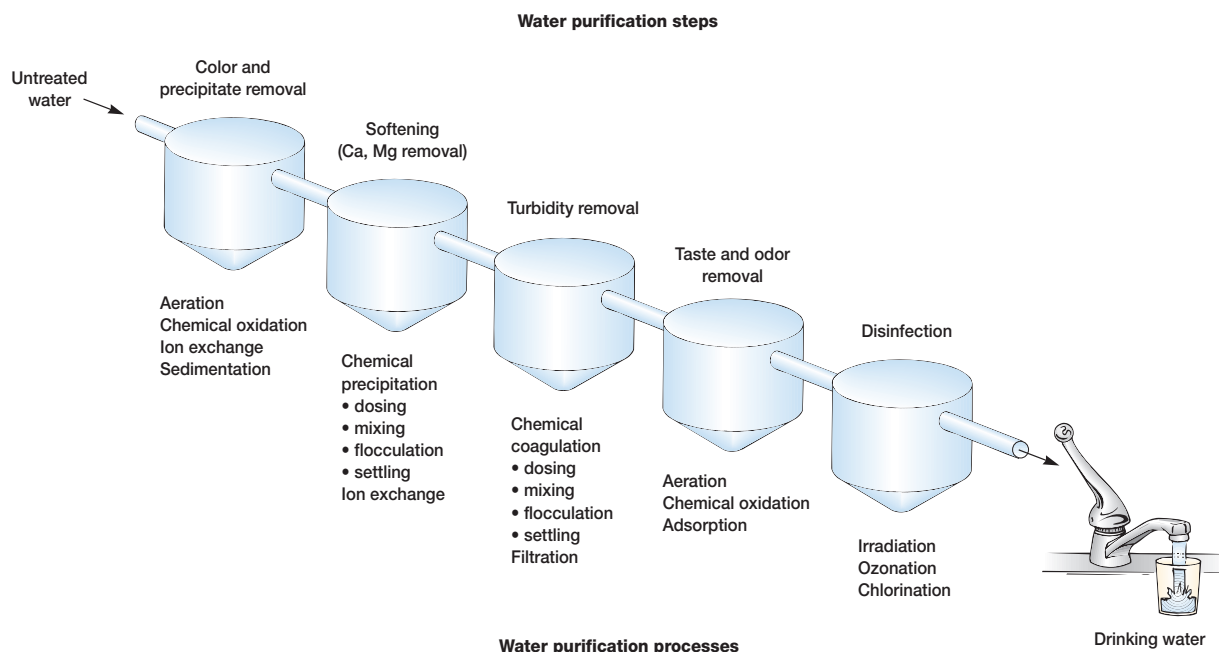


Figure 29.20 Water Purification. Several alternatives can be used for drinking water treatment depending on the initial water quality. A major concern is disinfection: chlorination can lead to the formation of disinfection byproducts (DBPs), including potentially carcinogenic trihalomethanes (THMs).

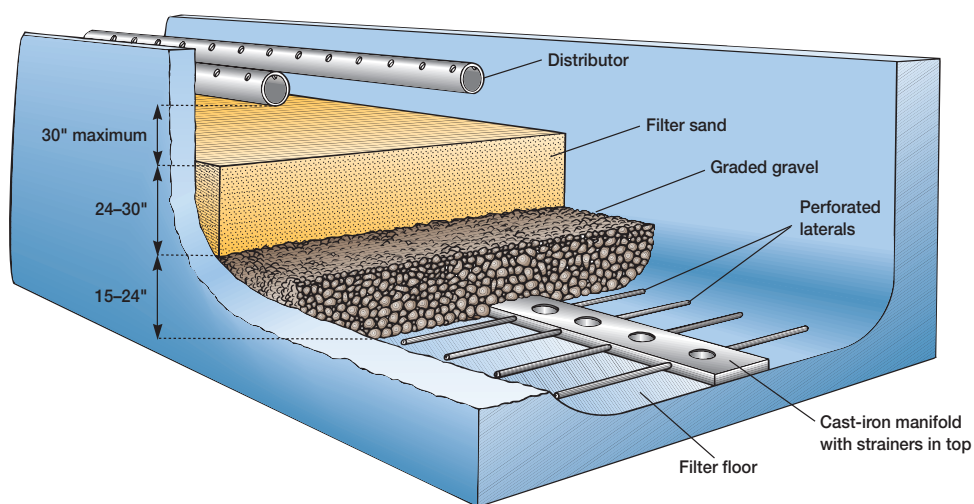


Figure 29.21 Water Filtration. Physical filtration is an important step in drinking water treatment. This is a cross section of a typical sand filter showing layers of sand and graded gravel.

steps. If the raw water contains a great deal of suspended material, it often is first routed to a **sedimentation basin** and held so that sand and other very large particles can settle out. The partially clarified water is then mixed with chemicals such as alum and lime and moved to a **settling basin** where more material pre-

cipitates out. This procedure is called **coagulation** or **flocculation** and removes microorganisms, organic matter, toxic contaminants, and suspended fine particles. After these steps the water is further purified by passing it through a filtration unit (**figure 29.21**). **Rapid sand filters**, which depend on the physical trap-

Box 29.2

Waterborne Diseases, Water Supplies, and Slow Sand Filtration: The Return of a Time-Tested Concept in Drinking Water Treatment

Slow sand filtration, in which drinking water is passed through a sand filter that develops a layer of microorganisms on its surface, has had a long and interesting history. After London's severe cholera epidemic of 1849, Parliament, in an act of 1852, required that the entire water supply of London be passed through slow sand filters before use.

The value of this process was shown in 1892, when a major cholera epidemic occurred in Hamburg, Germany, and 10,000 lives were lost. The neighboring town, Altona, which used slow sand filtration, did not

have a cholera epidemic. Slow sand filters were installed in many cities in the early 1900s, but the process fell into disfavor with the advent of rapid sand filters, chlorination, and the use of coagulants such as alum. Slow sand filtration, a time-tested process, is regaining favor because of its filtration effectiveness and lower maintenance costs. Slow sand filtration is particularly effective for the removal of *Giardia* from waters. For this reason slow sand filtration is used in many mountain communities where *Giardia* is a problem.

ping of fine particles and flocs are usually used for this purpose. This filtration removes up to 99% of the remaining bacteria. After filtration the water is treated with a disinfectant. This step usually involves chlorination, but ozonation is becoming increasingly popular. When chlorination is employed, the chlorine dose must be large enough to leave residual free chlorine at a concentration of 0.2 to 2.0 mg/liter. A concern is the creation of **disinfection by-products (DBPs)** such as **trihalomethanes (THMs)** that are formed when chlorine reacts with organic matter. Some of these compounds may be carcinogens.

The preceding purification process effectively removes or inactivates disease-causing bacteria and indicator organisms (coliforms). Unfortunately, however, the use of coagulants, rapid filtration, and chemical disinfection often does not consistently and reliably remove *Giardia lamblia* cysts, *Cryptosporidium* oocysts, *Cyclospora*, and viruses. *Giardia*, a cause of human diarrhea, is now recognized as the most common identified waterborne pathogen in the United States. The protozoan, first observed by Leeuwenhoek in 1681, has trophozoite and cyst forms. The disease often is called "backpacker's diarrhea" and is transmitted primarily through untreated stream water or undependable municipal water supplies. More consistent removal of *Giardia* cysts, which are about 7 to 10 by 8 to 12 μm in size, can be achieved with **slow sand filters**. This treatment involves the slow passage of water through a bed of sand in which a microbial layer covers the surface of each sand grain. Waterborne microorganisms are removed by adhesion to the gelatinous surface microbial layer (**Box 29.2**). *Giardia* (pp. 589, 953–54)

In the last few years, *Cryptosporidium* has become of even greater concern than *Giardia*. This protozoan parasite is smaller than *Giardia* and is even more difficult to remove from water. *Cryptosporidium* is discussed in section 40.2.

Another newly emerging protozoan human pathogen is *Cyclospora*. This coccidian protozoan is larger than *Cryptosporidium* and has two sporocysts, each with two sporozoites (*Cryptosporidium* has four sporozoites in its oocysts). *Cyclospora* causes cyclosporiasis, a self-limiting diarrhea that lasts from 19 to 43 days and can be accompanied by nausea, vomiting, cramps,

and fever. *Cyclospora* has recently been implicated in the occurrence of diarrheal diseases due to fecal contamination of imported fruits and vegetables.

Viruses in drinking water also must be destroyed or removed. Coagulation and filtration do reduce virus levels about 90 to 99%. Further inactivation of viruses by chemical oxidants, high pH, and photooxidation may yield a reduction as great as 99.9%. None of these processes, however, is considered to provide sufficient protection. New standards for virus inactivation are being developed. Bacteriophages (*see chapter 17*), which can be easily grown and assayed, now are being used as disinfection surrogates. If sufficient log reductions of bacteriophage infectivity occur with a given disinfection process, it is assumed that viruses capable of infecting humans will be reduced to satisfactory levels.

With the increased concern for drinking water quality in the United States, an Information Collection Rule (ICR) has been initiated. This program has been designed to assess the threat of these pathogens to the waters of cities with populations over 100,000.

1. Which important bacterial pathogens can be transmitted by waters?
2. What disease is caused by *Naegleria fowleri*? What is the route of entry to the human body?
3. What steps are usually taken to purify drinking water?
4. Why is chlorination, although beneficial in terms of bacterial pathogen control, of environmental concern?
5. Which important waterborne pathogens are not controlled reliably by chlorination?

Sanitary Analysis of Waters

Monitoring and detection of indicator and disease-causing microorganisms are a major part of sanitary microbiology. Bacteria from the intestinal tract generally do not survive in the aquatic environment, are under physiological stress, and gradually lose their ability to form colonies on differential and selective media.

Their die-out rate depends on the water temperature, the effects of sunlight, the populations of other bacteria present, and the chemical composition of the water. Procedures have been developed to attempt to “resuscitate” these stressed coliforms before they are identified using selective and differential media.

A wide range of viral, bacterial, and protozoan diseases result from the contamination of water with human fecal wastes (*see chapters 38, 39, and 40*). Although many of these pathogens can be detected directly, environmental microbiologists have generally used **indicator organisms** as an index of possible water contamination by human pathogens. Researchers are still searching for the “ideal” indicator organism to use in sanitary microbiology. The following are among the suggested criteria for such an indicator:

1. The indicator bacterium should be suitable for the analysis of all types of water: tap, river, ground, impounded, recreational, estuary, sea, and waste.
2. The indicator bacterium should be present whenever enteric pathogens are present.
3. The indicator bacterium should survive longer than the hardest enteric pathogen.
4. The indicator bacterium should not reproduce in the contaminated water and produce an inflated value.
5. The assay procedure for the indicator should have great specificity; in other words, other bacteria should not give positive results. In addition, the procedure should have high sensitivity and detect low levels of the indicator.
6. The testing method should be easy to perform.
7. The indicator should be harmless to humans.
8. The level of the indicator bacterium in contaminated water should have some direct relationship to the degree of fecal pollution.

Coliforms, including *Escherichia coli*, are members of the family *Enterobacteriaceae*. These bacteria make up approximately 10% of the intestinal microorganisms of humans and other animals (*see figure 31.2*) and have found widespread use as indicator organisms. They lose viability in fresh water at slower rates than most of the major intestinal bacterial pathogens. When such “foreign” enteric indicator bacteria are not detectable in a specific volume (100 ml) of water, the water is considered **potable** [Latin *potabilis*, fit to drink], or suitable for human consumption. *The Enterobacteriaceae* (pp. 505–7)

The coliform group includes *E. coli*, *Enterobacter aerogenes*, and *Klebsiella pneumoniae*. Coliforms are defined as facultatively anaerobic, gram-negative, nonsporing, rod-shaped bacteria that ferment lactose with gas formation within 48 hours at 35°C. The original test for coliforms that was used to meet this definition involved the presumptive, confirmed, and completed tests, as shown in **figure 29.22**. The presumptive step is carried out by means of tubes inoculated with three different sample volumes to give an estimate of the **most probable number (MPN)** of coliforms in the water. The complete process, including the confirmed and completed tests, requires at least 4 days of incubations and transfers.

Unfortunately the coliforms include a wide range of bacteria whose primary source may not be the intestinal tract. To deal with

this difficulty, tests have been developed that allow waters to be tested for the presence of **fecal coliforms**. These are coliforms derived from the intestine of warm-blooded animals, which can grow at the more restrictive temperature of 44.5°C.

To test for coliforms and fecal coliforms, and more effectively recover stressed coliforms, a variety of simpler and more specific tests have been developed. These include the membrane filtration technique, the **presence-absence (P-A) test** for coliforms and the related Colilert **defined substrate test** for detecting both coliforms and *E. coli*.

The **membrane filtration technique** (*see figure 6.6*), has become a common and often preferred method of evaluating the microbiological characteristics of water. The water sample is passed through a membrane filter. The filter with its trapped bacteria is transferred to the surface of a solid medium or to an absorptive pad containing the desired liquid medium. Use of the proper medium allows the rapid detection of total coliforms, fecal coliforms, or fecal streptococci by the presence of their characteristic colonies (**figure 29.23**; *see also figure 6.7*). Samples can be placed on a less selective resuscitation medium, or incubated at a less stressful temperature, prior to growth under the final set of selective conditions. An example of a resuscitation step is the use of a 2 hour incubation on a pad soaked with lauryl sulfate broth, as is carried out in the LES Endo procedure. A resuscitation step often is needed with chlorinated samples, where the microorganisms are especially stressed. The advantages and disadvantages of the membrane filter technique are summarized in **table 29.4**. Membrane filters have been widely used with water that does not contain high levels of background organisms, sediment, or heavy metals.

More simplified tests for detecting coliforms and fecal coliforms are now available. The presence-absence test (P-A test) can be used for coliforms. This is a modification of the MPN procedure, in which a larger water sample (100 ml) is incubated in a single culture bottle with a triple-strength broth containing lactose broth, lauryl tryptose broth, and bromocresol purple indicator. The P-A test is based on the assumption that no coliforms should be present in 100 ml of drinking water. A positive test results in the production of acid (a yellow color) and constitutes a positive presumptive test requiring confirmation.

To test for both coliforms and *E. coli*, the related Colilert defined substrate test can be used. A water sample of 100 ml is added to a specialized medium containing *o*-nitrophenyl- β -D-galactopyranoside (**ONPG**) and 4-methylumbelliferyl- β -D-glucuronide (**MUG**) as the only nutrients. If coliforms are present, the medium will turn yellow within 24 hours at 35°C due to the hydrolysis of ONPG, which releases *o*-nitrophenol, as shown in **figure 29.24**. To check for *E. coli*, the medium is observed under long-wavelength UV light for fluorescence. When *E. coli* is present, the MUG is modified to yield a fluorescent product. If the test is negative for the presence of coliforms, the water is considered acceptable for human consumption. The main change from previous standards is the requirement to have waters free of coliforms and fecal coliforms. If coliforms are present, fecal coliforms or *E. coli* must be tested for.

Molecular techniques are now used routinely to detect coliforms in waters and other environments, including foods.

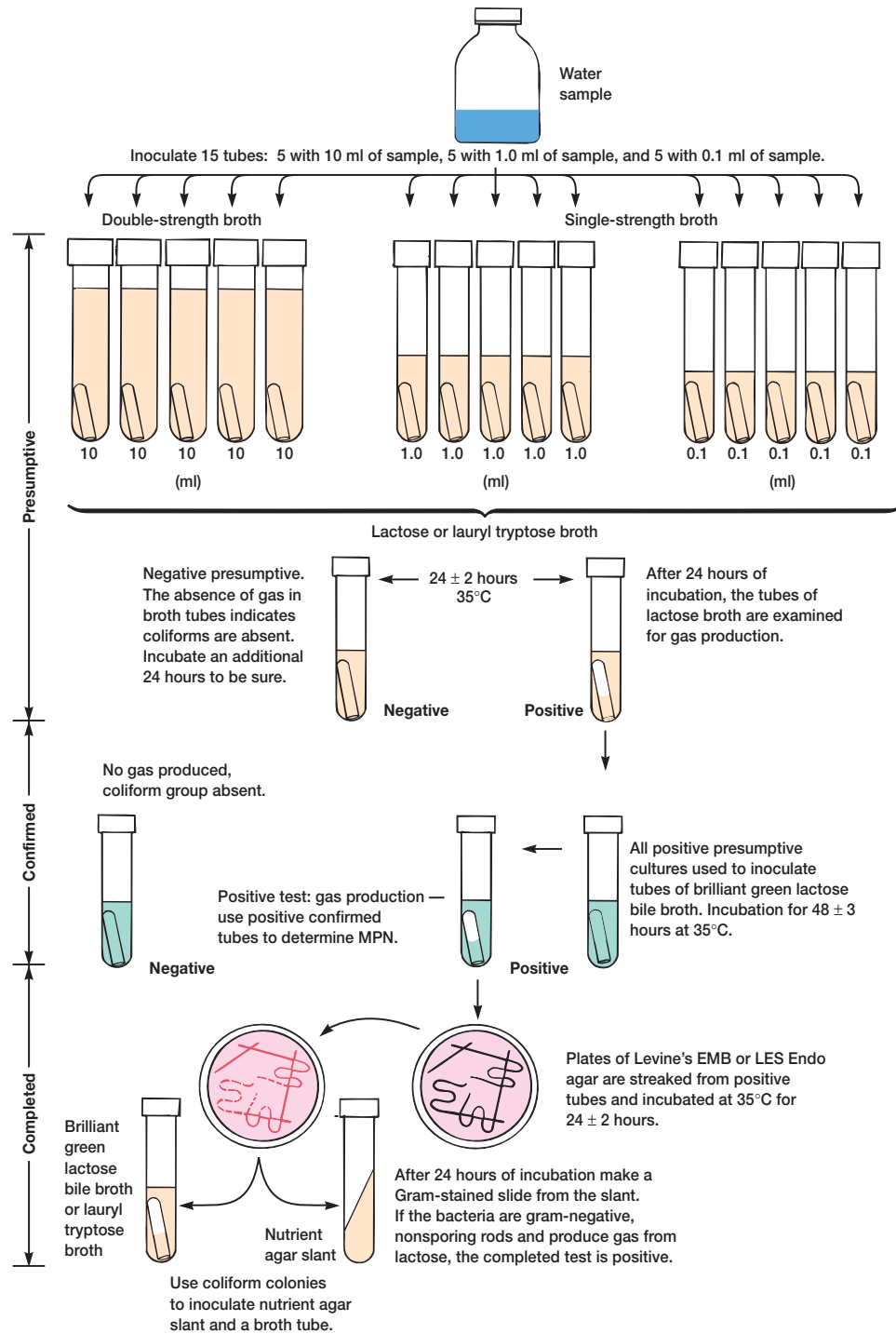


Figure 29.22 The Multiple-Tube Fermentation Test. The multiple-tube fermentation technique has been used for many years for the sanitary analysis of water. Lactose broth tubes are inoculated with different water volumes in the presumptive test. Tubes that are positive for gas production are inoculated into brilliant green lactose bile broth in the confirmed test, and positive tubes are used to calculate the most probable number (MPN) value. The completed test is used to establish that coliform bacteria are present.

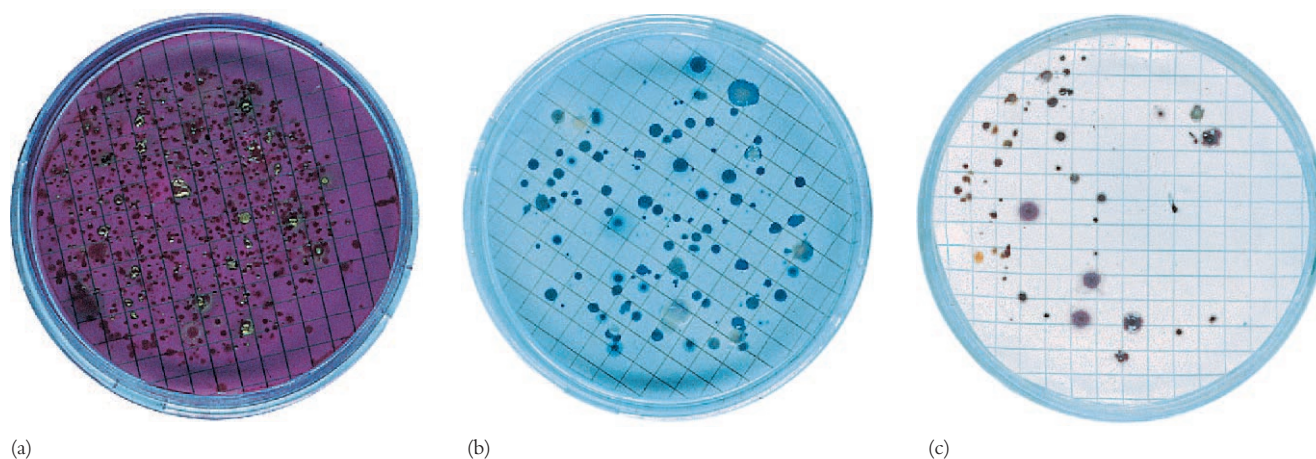


Figure 29.23 Coliform and Enterococcal Colonies. Membrane filters made it possible to more rapidly test waters for the presence of coliforms, fecal coliforms, and fecal enterococci by the use of differential media. (a) Coliform reactions on an Endo medium. (b) Fecal coliform growth on a bile salt medium (m-FC agar) containing aniline blue dye. (c) Fecal enterococci growing on an azide-containing medium (KF agar) with TTC, an artificial electron acceptor, added to allow better detection of colonies.

Table 29.4 Advantages and Disadvantages of the Membrane Filter Technique for Evaluation of the Microbial Quality of Water

Advantages

- Good reproducibility
- Single-step results often possible
- Filters can be transferred between different media
- Large volumes can be processed to increase assay sensitivity
- Time savings are considerable
- Ability to complete filtrations on site
- Lower total cost in comparison with MPN procedure

Disadvantages

- High-turbidity waters limit volumes sampled
- High populations of background bacteria cause overgrowth
- Metals and phenols can adsorb to filters and inhibit growth

Source: Data from A. E. Greenberg, et al., *Standard Methods for the Examination of Water and Wastewater*, 16th edition, page 886, 1985. American Public Health Association, Washington, D.C.



Figure 29.24 The Defined Substrate Test. This much simpler test is now being used to detect coliforms and fecal coliforms in single 100 ml water samples. The medium uses ONPG and MUG (see text) as defined substrates. (a) Uninoculated control. (b) Yellow color due to the presence of coliforms. (c) Fluorescent reaction due to the presence of fecal coliforms.

16 S rRNA gene-targeted primers for coliforms have been developed. Using these primers, it is possible to detect one colony-forming unit (CFU) of *E. coli* per 100 ml of water, if an eight-hour enrichment step precedes the use of the PCR amplification. This allows the differentiation of nonpathogenic and enterotoxigenic strains, including the shiga-toxin producing *E. coli* O157:H7. The PCR technique (pp. 326–27)

In the United States a set of general guidelines for microbiological quality of drinking waters has been developed, including standards for coliforms, viruses, and *Giardia* (table 29.5). If unfiltered surface waters are being used, one coliform test must be run each day when the waters have higher turbidities.

Other indicator microorganisms include **fecal enterococci**. The fecal enterococci are increasingly being used as an indicator

Table 29.5 Current Drinking Water Standards in the United States.

Agent	Allowable Maximum Contaminant Level Goal (MCLG) or Maximum Contaminant Level (MCL)
Coliforms	MCLG = 0 MCL = No more than 5% positive total coliform samples/month for water systems that collect > 40 samples/month. For water systems that collect < 40 routine samples/month, no more than 1 can be coliform positive. Every sample that has total coliforms must be analyzed for fecal coliforms. There cannot be any fecal coliforms.
<i>Giardia lamblia</i>	MCLG = 0
<i>Legionella</i>	MCLG = 0
Viruses (enteric)	MCLG = 0

Source: Environmental Protection Agency, USA, 24 July, 2000.

of fecal contamination in brackish and marine water. In salt water these bacteria die back at a slower rate than the fecal coliforms, providing a more reliable indicator of possible recent pollution. [The genus *Enterococcus* \(pp. 531–33\)](#)

1. What is an indicator organism, and what properties should it have?
2. How is a coliform defined? How does this definition relate to presumptive, confirmed, and completed tests?
3. How does one differentiate between coliforms and fecal coliforms in the laboratory?
4. In what type of environment is it better to use fecal enterococci rather than fecal coliforms as an indicator organism?
5. What are the advantages and disadvantages of membrane filters for microbiological examinations of water?
6. Why has the defined substrate test with ONPG and MUG been accepted as a test of drinking water quality?

29.6 Wastewater Treatment

Waters often contain high levels of organic matter from industrial and agricultural wastes (e.g., from food processing, petrochemical and chemical plants, and plywood plant resin wastes), and from human wastes. It is necessary to remove organic matter by the process of wastewater treatment. Depending on the effort given to this task, it may still produce waters containing nutrients and some microorganisms, which can be released to rivers and streams.

The treatment of human waste-impacted waters, with the many pathogens that may be present, is one of the most important factors in maintaining an advanced society. In fact, the use of modern sewage treatment, together with chlorination, has led to major reductions in the worldwide spread of pathogens. This often unappreciated technology thus is at the front line of main-

Table 29.6 The Biochemical Oxygen Demand (BOD) Test: A System with Excess and Limiting Components

Components in Excess at the End of the Incubation Period

Nitrogen
Phosphorus
Iron
Trace elements
Microorganisms
Oxygen

Component Limiting at the End of the Incubation Period

Organic matter

taining a healthy society. When sewage treatment systems break down, as often occurs with natural disasters or civil unrest, long-forgotten diseases such as cholera reappear.

Measuring Water Quality

To measure carbon removal in these wastewater treatment processes, several approaches can be used. Carbon removal can be measured (1) as **total organic carbon (TOC)**, (2) as chemically oxidizable carbon by the **chemical oxygen demand (COD)** test, or (3) as biologically usable carbon by the **biochemical oxygen demand (BOD)** test. The TOC includes all carbon, whether or not it is usable by microorganisms. This is carried out by oxidizing the organic matter in a sample at high temperature in an oxygen stream and measuring the resultant CO₂ by infrared or potentiometric techniques. The COD gives a similar measurement, except that lignin often will not react with the oxidizing chemical, such as permanganate, that is used in this procedure. The BOD test, in comparison, measures only the portion of the total carbon that can be oxidized by microorganisms in a 5-day period under standard conditions.

The biochemical oxygen demand is an indirect measure of organic matter in aquatic environments. It is the amount of dissolved O₂ needed for microbial oxidation of biodegradable organic matter. When O₂ consumption is measured, the O₂ itself must be present in excess and not limit oxidation of the nutrients (**table 29.6**). To achieve this, the waste sample is diluted to assure that at least 2 mg/liter of O₂ are used while at least 1 mg/liter of O₂ remains in the test bottle. Ammonia released during organic matter oxidation can also exert an O₂ demand in the BOD test, so nitrification or the **nitrogen oxygen demand (NOD)** is often inhibited by 2-chloro-6-(trichloromethyl) pyridine (nitrapyrin). In the normal BOD test, which is run for 5 days at 20°C on untreated samples, nitrification is not a major concern. However, when treated effluents are analyzed, NOD can be a problem.

In terms of speed, the TOC is fastest, but less informative in terms of biological processes. The COD is slower and involves the use of wet chemicals with higher waste chemical disposal costs. The TOC, COD, and BOD provide different but

Table 29.7 Major Steps in Primary, Secondary, and Tertiary Treatment of Wastes

Treatment Step	Processes
Primary	Removal of insoluble particulate materials by settling, screening, addition of alum and other coagulation agents, and other physical procedures
Secondary	Biological removal of dissolved organic matter Trickling filters Activated sludge Lagoons Extended aeration systems Anaerobic digesters
Tertiary	Biological removal of inorganic nutrients Chemical removal of inorganic nutrients Virus removal/inactivation Trace chemical removal

complementary information on the carbon in a water sample. It is critical to note that these measurements, concerned with carbon and carbon removal, do not directly address concerns for removal of minerals such as nitrate, phosphate, and sulfate from waters. These minerals are having worldwide impacts on cyanobacterial and algal growth in lakes, rivers, and the oceans by contributing to the process of eutrophication. The removal of dissolved organic matter and possibly inorganic nutrients, plus inactivation and removal of pathogens, are important parts of wastewater treatment.

1. What are TOC, COD, and BOD and how are these similar and different?
2. What factors can lead to a nitrogen oxygen demand (NOD) in a water?
3. What components should limit the reactions in a BOD test, and what components should not limit reaction rates? Why?
4. What minerals can contribute to eutrophication?

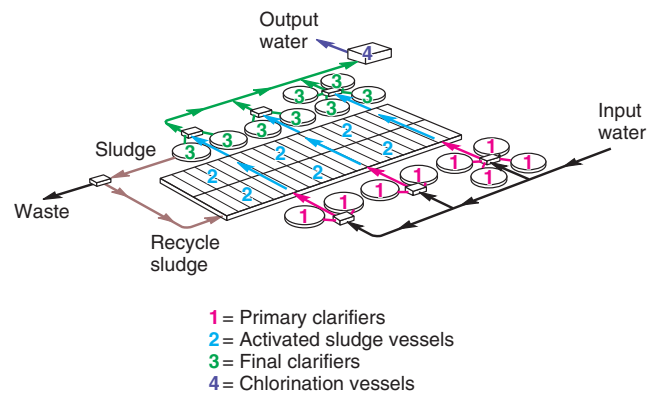
Water Treatment Processes

The aerobic self-purification sequence that occurs when organic matter is added to lakes and rivers can be carried out under controlled conditions in which natural processes are intensified. This often involves the use of large basins (conventional sewage treatment) where mixing and gas exchange are carefully controlled. Purification also can be carried out using constructed wetlands where natural reed and aquatic plant communities (and their associated microorganisms) process dissolved nutrients. Such procedures can be effective in minimizing environmental degradation of valuable water resources and in destroying potential human pathogens.

Conventional **wastewater treatment** normally involves primary, secondary, and tertiary treatment, as summarized in **table 29.7**.



(a)



(b)

Figure 29.25 An Aerial View of a Modern Conventional Sewage Treatment Plant. Sewage treatment plants allow natural processes of self-purification that occur in rivers and lakes to be carried out under more intense, managed conditions in large concrete vessels. (a) A plant in New Jersey. (b) A diagram of flows in the plant.

Primary treatment can physically remove 20 to 30% of the BOD that is present in particulate form. In this treatment, particulate material is removed by screening, precipitation of small particulates, and settling in basins or tanks. The resulting solid material is usually called **sludge**.

An aerial photograph of a modern sewage treatment plant is shown in **figure 29.25a**, and a schematic of this plant is shown in **figure 29.25b**. The sequence of physical removal of settleable solids (primary treatment) and secondary treatment (biological transformation of dissolved organic matter to microbial biomass and carbon dioxide) is shown, together with the final clarifiers. The final clarifiers separate the newly formed microbial biomass (sewage sludge) from the processed water stream, which can be returned to the receiving body of water. At the end of the process, the water is usually chlorinated (itself an emerging environmental and human health problem) before it is released.

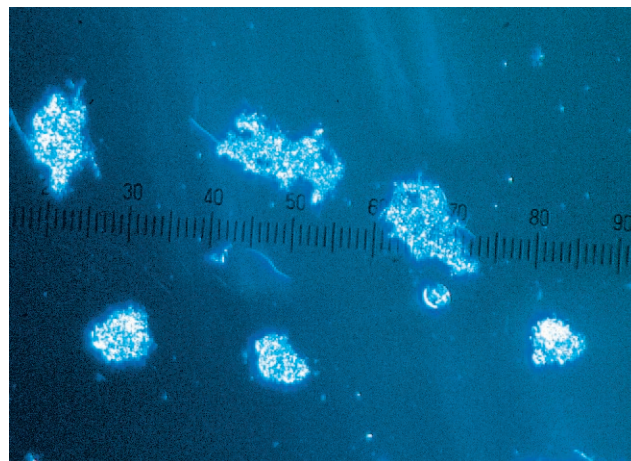
Secondary treatment is used after primary treatment for the biological removal of dissolved organic matter. About 90 to 95% of the BOD and many bacterial pathogens are removed by this process. Several approaches can be used in secondary treatment to biologically remove dissolved organic matter. All of these techniques involve similar microbial activities. Under aerobic conditions, dissolved organic matter will be transformed into additional microbial biomass plus carbon dioxide. When microbial growth is completed, under ideal conditions the microorganisms will aggregate and form a settleable stable floc structure. Minerals in the water also may be tied up in microbial biomass. When microorganisms grow, flocs can form (**figure 29.26**). As shown in figure 29.26a, a healthy settleable floc will be compact. In contrast, poorly formed floc can have a network of filamentous microbes that will retard settling (figure 29.26b).

When these processes occur with lower O_2 levels or with a microbial community that is too young or too old, unsatisfactory floc formation and settling can occur. The result is a **bulking sludge**, caused by the massive development of filamentous bacteria such as *Sphaerotilus* and *Thiothrix*, together with many poorly characterized filamentous organisms. These important filamentous bacteria (see pp. 496, 502) form flocs that do not settle well, and thus produce effluent quality problems.

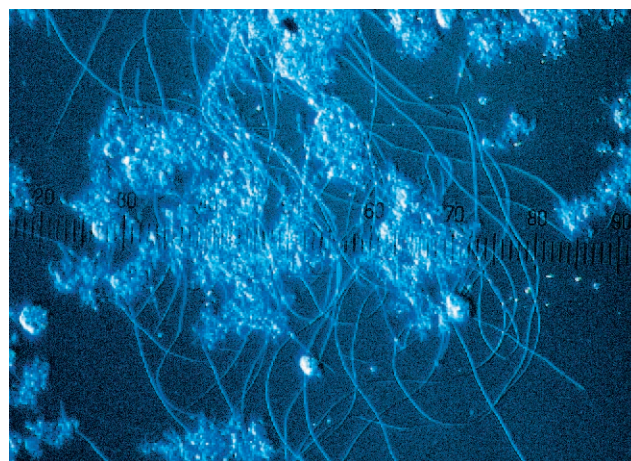
An aerobic **activated sludge** system (**figure 29.27a**) involves a horizontal flow of materials with a recycle of sludge—the active biomass that is formed when organic matter is oxidized and degraded by microorganisms. Activated sludge systems can be designed with variations in mixing. In addition, the ratio of organic matter added to the active microbial biomass can be varied. A low-rate system (low nutrient input per unit of microbial biomass), with slower growing microorganisms, will produce an effluent with low residual levels of dissolved organic matter. A high-rate system (high nutrient input per unit of microbial biomass), with faster growing microorganisms, will remove more dissolved organic carbon per unit time, but produce a poorer quality effluent.

Aerobic secondary treatment also can be carried out with a **trickling filter** (**figure 29.27b**). The waste effluent is passed over rocks or other solid materials upon which microbial films have developed, and the microbial community in these films degrades the organic waste. A sewage treatment plant can be operated to produce less sludge by employing the **extended aeration** process (**figure 29.27c**). Microorganisms grow on the dissolved organic matter, and the newly formed microbial biomass is eventually consumed to meet maintenance energy requirements. This requires extremely large aeration basins and extended aeration times. In addition, with the biological self-utilization of the biomass, minerals originally present in the microorganisms are again released to the water.

All aerobic processes produce excess microbial biomass, or sewage sludge, which contains many recalcitrant organics. Often the sludges from aerobic sewage treatment, together with the materials settled out in primary treatment, are further treated by **anaerobic digestion** (**figure 29.28**). Anaerobic digesters are



(a)



(b)

Figure 29.26 Proper Floc Formation in Activated Sludge.

Microorganisms play a critical role in the functioning of activated sludge systems. The operation is dependent on the formation of settleable flocs (**a**). If the plant does not run properly, poorly settling flocs (**b**) can form due to such causes as low aeration, sulfide, and acidic organic substrates. These flocs do not settle properly because of their open or porous structure. As a consequence, the organic material is released with the treated water and lowers the quality of the final effluent.

large tanks designed to operate anaerobically with continuous input of untreated sludge and removal of the final, stabilized sludge product. Methane is vented and often burned for heat and electricity production. This digestion process involves three steps: (1) the fermentation of the sludge components to form organic acids, including acetate; (2) production of the methanogenic substrates: acetate, CO_2 , and hydrogen; and finally, (3) methanogenesis by the methane producers. These methanogenic processes,

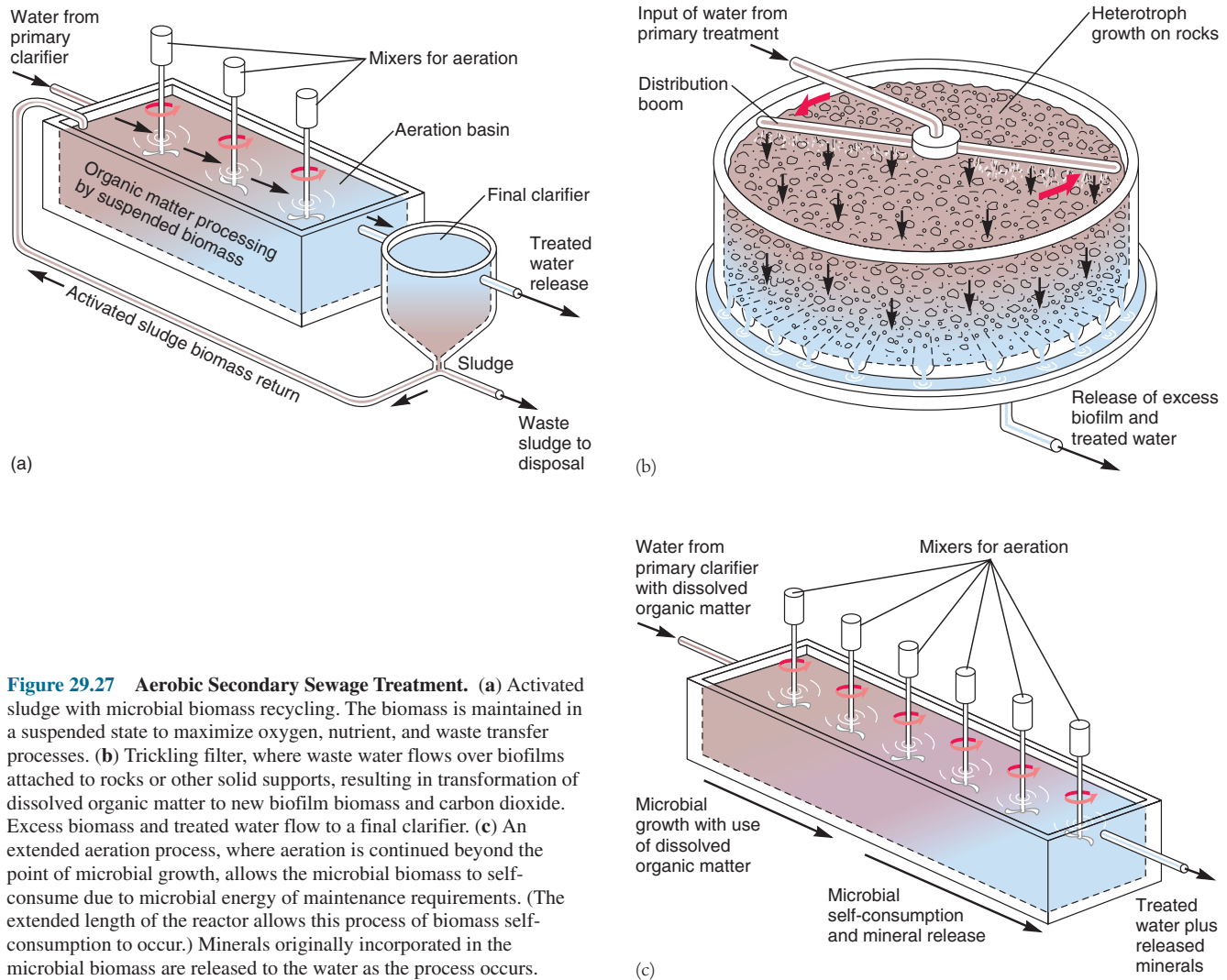


Figure 29.27 Aerobic Secondary Sewage Treatment. (a) Activated sludge with microbial biomass recycling. The biomass is maintained in a suspended state to maximize oxygen, nutrient, and waste transfer processes. (b) Trickling filter, where waste water flows over biofilms attached to rocks or other solid supports, resulting in transformation of dissolved organic matter to new biofilm biomass and carbon dioxide. Excess biomass and treated water flow to a final clarifier. (c) An extended aeration process, where aeration is continued beyond the point of microbial growth, allows the microbial biomass to self-consume due to microbial energy of maintenance requirements. (The extended length of the reactor allows this process of biomass self-consumption to occur.) Minerals originally incorporated in the microbial biomass are released to the water as the process occurs.



Figure 29.28 High-Efficiency Anaerobic Bioreactors Used for Sludge Digestion and Methane Production. These “egg-shaped” units are so well insulated that they maintain their temperature (39°C) without outside heating, and the methane can be burned to provide electricity. These reactors are located at Kiel, Germany.

Box 29.3

Sewage Sludge, Long-Term Concerns with Land and Water Disposal: An Environmental Challenge

Sewage treatment plants have allowed large cities to develop near rivers and lakes and still maintain the quality of the water. Large quantities of sludge are produced during sewage treatment and are usually subjected to anaerobic digestion. This process transforms complex organic matter (including microorganisms that grew in the aerobic treatment process) to methane and CO₂. At the same time heavy metals are concentrated in the residual sludge, and viable cysts of free-living protozoa may be present.

The sludge is disposed of on land or, in large urban areas near the ocean, by dumping at designated disposal sites in coastal waters. When

such sludges are dumped offshore, free-living protozoa of the genus *Acanthamoeba* may be released in the water, where they may infect bathers. *Acanthamoeba* infections are more widespread than usually is recognized, and the organisms can be common contaminants in water that tests negative for coliforms and fecal coliforms.

The use of water-based waste disposal systems has, of course, allowed major improvements in urban living and the retirement of chamber pots to museums. A product of this advancement in public health is sewage sludge. NIMBY (not in my backyard) has been the usual means of managing this problem.

Table 29.8 Sequential Reactions in the Anaerobic Biological Use of Organic Wastes

Process Step	Substrates	Products	Major Microorganisms
Fermentation	Organic polymers	Butyrate, propionate, lactate, succinate, ethanol, acetate, ^a H ₂ , ^a CO ₂ ^a	<i>Clostridium</i> <i>Bacteroides</i> <i>Peptostreptococcus</i> <i>Peptococcus</i> <i>Eubacterium</i> <i>Lactobacillus</i>
Acetogenic reactions	Butyrate, propionate, lactate, succinate, ethanol	Acetate, H ₂ , CO ₂	<i>Syntrophomonas</i> <i>Syntrophobacter</i> <i>Acetobacterium</i>
Methanogenic reactions	Acetate	CH ₄ + CO ₂	<i>Methanosarcina</i> <i>Methanotrix</i>
	H ₂ and HCO ₃ ⁻	CH ₄	<i>Methanobrevibacter</i> <i>Methanomicrobium</i> <i>Methanogenium</i> <i>Methanobacterium</i> <i>Methanococcus</i> <i>Methanospirillum</i>

^aMethanogenic substrates produced in the initial fermentation step.

summarized in **table 29.8**, involve critical balances between oxidants and reductants. To function most efficiently, the hydrogen concentration must be maintained at a low level. If hydrogen and organic acids accumulate, methane production can be inhibited, resulting in a stuck digester. [Methanogenic archaea \(pp. 458–61\)](#)

Anaerobic digestion has many advantages. Most of the microbial biomass produced in aerobic growth is used for methane production. The resulting sludge occupies less volume and can be dried easily. However, heavy metals and other environmental contaminants are often concentrated in the sludge. There may be longer-term environmental and public health effects from disposal of this material on land or in water (**Box 29.3**).

The sludge resulting from anaerobic digestion, instead of being disposed of on soil or added to waters, can be further treated under aerobic conditions in the sewage treatment plant. This removes more pathogens and oxidizes smelly ammonium and sulfide to odorless oxidized forms. The combined process takes advantage of the complementary nature of microbial growth under sequential anaerobic-aerobic conditions.

Tertiary treatment purifies wastewaters more than is possible with primary and secondary treatments. The goal is to remove such pollutants as nonbiodegradable organic material (e.g., polychlorinated biphenyls), heavy metals, and minerals. It is particularly important to remove nitrogen and phosphorus compounds

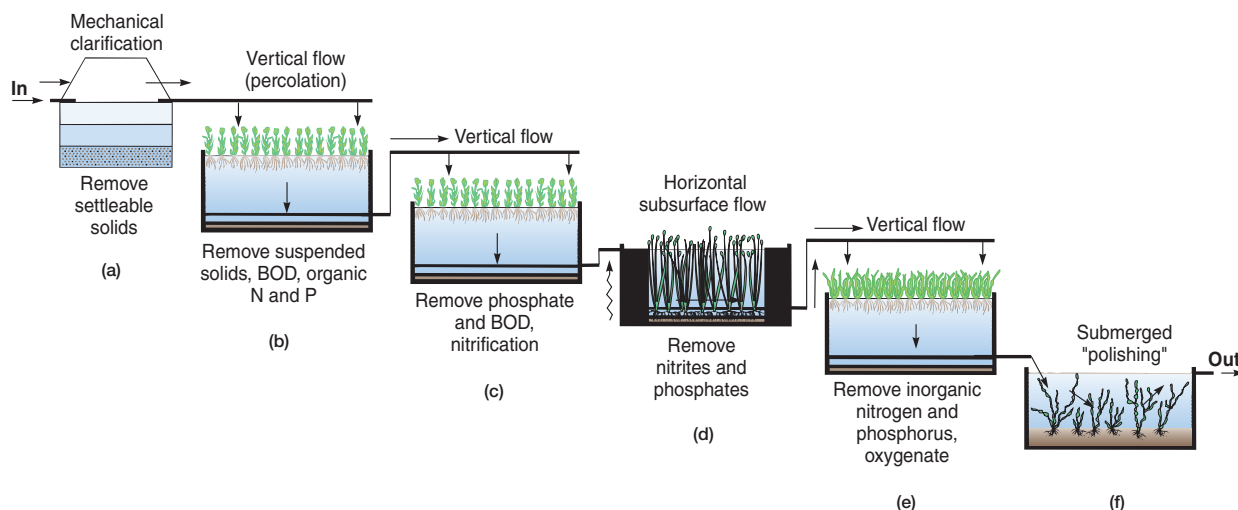


Figure 29.29 **Constructed Wetland for Wastewater Treatment.** Multistage constructed wetland systems can be used for organic matter and phosphate removal. Free-floating macrophytes (**b,c,e**), such as duckweed and water hyacinth, can be used for a variety of purposes. Emergent macrophytes (**d**) such as bulrush, allow surface flow as well as vertical and horizontal subsurface flow. Submerged vegetation (**f**), such as waterweed, allows final “polishing” of the water. These wetlands also can be designed for nitrification and metal removal from waters.

that can promote eutrophication. Organic pollutants can be removed with activated carbon filters. Phosphate usually is precipitated as calcium or iron phosphate (for example, by the addition of lime). Excess nitrogen may be removed by “stripping,” volatilization as NH_3 at high pHs. Ammonia itself can be chlorinated to form dichloramine, which is then converted to molecular nitrogen. In some cases, biological processes also are used to remove nitrogen and phosphorus. A widely used process for nitrogen removal is denitrification (*see p. 190*). Here nitrate, produced under aerobic conditions, is used as an electron acceptor under conditions of low oxygen with organic matter added as an energy source. Nitrate reduction yields nitrogen gas and nitrous oxide (N_2O) as the major products. For phosphorus removal, aerobic and anaerobic conditions are used alternately in a series of treatments, and phosphorus accumulates in specially adapted microbial biomass as polyphosphate. Tertiary treatment is expensive and usually not employed except where necessary to prevent obvious ecological disruption.

Anaerobic and aerobic treatment processes often are used together in a carefully designed sequence as a part of tertiary treatment. Linked aerobic and anaerobic processes now are designated by the acronym AAO, which means anaerobic-anoxic-oxic. The complete sequence has three stages: (1) the anaerobic (A) processing of waste, (2) treatment of this product with added nitrate under anoxic conditions (A) to promote denitrification, and (3) “polishing” the effluent under aerobic (oxic) conditions (O) before release to the environment.

Wetlands are a vital natural resource and a critical part of our environment, and increasingly efforts are being made to

protect these fragile aquatic communities from pollution. Surprisingly a major means of treatment of wastes is the use of **constructed wetlands** where the basic components of natural wetlands (soils, aquatic plants, waters) are used as a functional waste treatment system. Constructed wetlands now are increasingly employed in the treatment of liquid wastes and for bioremediation (*see pp. 1012–14*). This system uses floating, emergent, or submerged plants, as shown in **figure 29.29**. The aquatic plants provide nutrients in the root zone, which can support microbial growth. Especially with the emergent plants, the root zone can be maintained in an anaerobic state in which sulfide, synthesized by *Desulfovibrio* using root zone organic matter as an energy source, can trap metals. Such emergent plant systems have found wide applications in the processing of waters from abandoned mines.

The different aquatic plant types and their associated microorganisms can be used in integrated systems to remove organic matter, inorganic nutrients, and metals from waters. Constructed wetlands also are being used to treat acid mine drainage (AMD) in many parts of the world. Higher-strength industrial wastes also can be treated.

Another method of effluent processing is simple surface flow soil treatment. In this approach liquid waste is allowed to flow across a planted or plowed field, where aerobic microbial processing of the waste can occur. This is becoming a widespread practice, with the increases in confined animal agriculture that are occurring in many areas. Whether dealing with beef cattle, hogs, or poultry, these large commercial operations produce massive pollution, especially when these facilities are placed near estuarine and coastal areas.

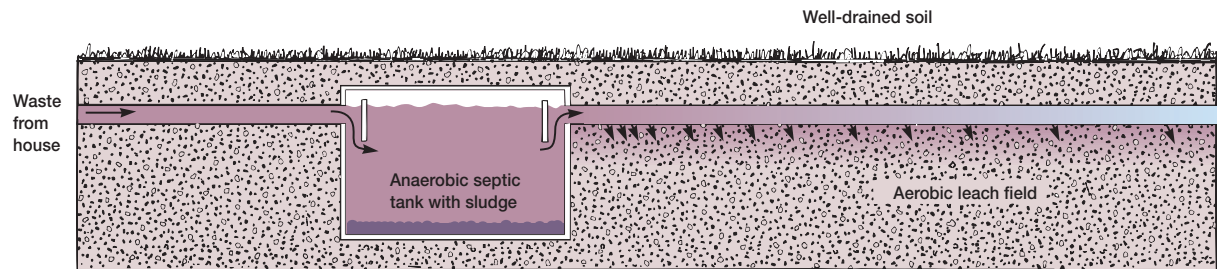


Figure 29.30 The Septic Tank Home Treatment System. This system combines an anaerobic waste liquefaction unit (the septic tank) with an aerobic leach field. Biological oxidation of the liquefied waste takes place in the leach field, unless the soil becomes flooded.

1. Explain how primary, secondary, and tertiary treatment are accomplished.
2. What is bulking sludge? Name several important microbial groups that contribute to this problem.
3. What are the steps of organic matter processing that occur in anaerobic digestion? Why is acetogenesis such an important step?
4. After anaerobic digestion is completed, why is sludge disposal still of concern? How can it be further treated to improve its quality?
5. Why might different aquatic plant types be used in constructed wetlands?

29.7 Groundwater Quality and Home Treatment Systems

Groundwater, or water in gravel beds and fractured rocks below the surface soil, is a widely used but often unappreciated water resource. In the United States groundwater supplies at least 100 million people with drinking water, and in rural and suburban areas beyond municipal water distribution systems, 90 to 95% of all drinking water comes from this source.

The great dependence on this resource has not resulted in a corresponding understanding of microorganisms and microbiological processes that occur in the groundwater environment. Increasing attention is now being given to predicting the fate and effects of groundwater contamination on the chemical and microbiological quality of this resource. Pathogenic microorganisms and dissolved organic matter are removed from water during subsurface passage through adsorption and trapping by fine sandy materials, clays, and organic matter. Microorganisms associated with these materials—including predators such as protozoa—can use the trapped pathogens as food. This results in purified water with a lower microbial population.

This combination of adsorption-biological predation is used in home treatment systems (figure 29.30). Conventional **septic tank** systems include an anaerobic liquefaction and digestion step that occurs in the septic tank itself (the tank functions as a simple anaerobic digester). This is followed by organic matter adsorption

and entrapment of microorganisms in the aerobic leach-field environment where biological oxidation occurs. A septic tank may not operate correctly for several reasons. It will not function properly if the retention time of the waste in the septic tank is too short. Retention time decreases when the flow is too rapid or when excessive sludge has accumulated in the septic tank. As a result undigested solids move into the leach field, gradually plugging the system. If the leach field floods and becomes anaerobic, biological oxidation does not occur, and effective treatment ceases.

When a suitable soil is not present and the septic tank outflow drains too rapidly to the deeper subsurface, problems can occur. Fractured rocks and coarse gravel materials provide little effective adsorption or filtration. This may result in the contamination of well water with pathogens and the transmission of disease. In addition, phosphorus from the waste will not be retained effectively and may pollute the groundwater. This often leads to nutrient enrichment of ponds, lakes, and rivers as the subsurface water enters these environmentally sensitive water bodies.

Subsurface zones (*see section 30.8*) also can become contaminated with pollutants from other sources. Land disposal of sewage sludges, illegal dumping of septic tank pumpage, improper toxic waste disposal, and runoff from agricultural operations all contribute to groundwater contamination with chemicals and microorganisms. Deep-well injection of industrial wastes has raised questions about the longer-term fate and effects of these materials.

Many pollutants that reach the subsurface will persist and may affect the quality of groundwater for extended periods. Much research is being conducted to find ways to treat groundwater in place—in **situ treatment**. Microorganisms and microbial processes are critical in many of these remediation efforts (*see chapter 42*).

1. In rural areas, approximately what percentage of the water used for human consumption is groundwater?
2. What factors can limit microbial activity in subsurface environments? Consider the energetic and nutritional requirements of microorganisms in your answer.
3. How, in principle, are a septic tank system and a leach-field system supposed to work? What factors can reduce the effectiveness of this system?

Summary

1. The largest portion of waters on the Earth is marine (97%). Most of this is cold (2 to 3°C) and at high pressure. Fresh waters are a minor but important part of the Earth's waters.
2. The movement of materials in aquatic environments extends over long distances and occurs at varied rates. Mixing and diffusion are critical processes in creating unique environments for different microorganisms.
3. Oxygen solubility and diffusion rates in waters are limited; waters are low oxygen diffusion rate environments, in comparison with soils (**figure 29.1**). Carbon dioxide, nitrogen, hydrogen, and methane are also important gases for microbial functioning in waters.
4. If sufficient organic matter is present, heterotrophic microorganisms can use this dissolved O₂ more rapidly than it can be replenished. Respiration or decomposition of algal and cyanobacterial blooms also can cause O₂ depletion.
5. Nutrient concentrations in waters can vary, and diffusion produces unique environments for microorganisms by creating gradients. The Winogradsky column is a model laboratory system that allows gradients and unique microbial communities to develop (**figure 29.3**).
6. The nutrient composition of the ocean influences the C:N:P ratio of the phytoplankton, which is called the Redfield ratio. This ratio is important for predicting nutrient cycling in oceans. Atmospheric additions of minerals, including iron and nitrogen, affect this ratio and global-level oceanic processes.
7. The microbial loop is an important part of the food web in waters (**figure 29.4**). This results from bacterial growth on organic matter released by the phytoplankton, followed by use of the bacteria as a prey by protozoa. The nutrients in a portion of these bacterial prey are then released in mineral forms, making them again available for use by the phytoplankton.
8. The marine microbial community is dominated, in terms of numbers and biomass, by ultramicrobacteria or nanobacteria. Archaea are important components of the microbial community. Viruses are present at high concentrations in many waters, and occur at 10-fold higher levels than the bacteria. In marine systems they may play a major role in controlling cyanobacterial development and nutrient turnover.
9. Many unusual microbial groups are found in waters, especially when oxidants and reductants can be linked. These include *Thioploca* and *Thiomargarita*, both of which are found in coastal areas where nutrient mixing occurs. *Thiomargarita* is the world's largest known bacterium (**figure 29.5**).
10. Aquatic fungi are important parts of the aquatic microbial community. These include the chytrids (**figure 29.7**), with a motile zoospore stage, and the Ingoldian fungi (**figure 29.8**), which often have tetradiate structures. Both of these are uniquely adapted to an aquatic existence, and the chytrids may contribute to disease in amphibians.
11. Microorganisms and their metabolic products interact with the atmosphere and are important in the field of aerobiology.
12. Barophiles are important in deep marine environments. These include barotolerant, barophilic, and extreme barophilic procaryotes.
13. Materials can move in aquatic environments over extended distances and depths. This may include animal and human wastes, carcasses, and plant detritus. Dust is transported over wide distances. All of these processes result in the constant introduction of new organisms.
14. The global cycles of carbon, nitrogen, phosphorus and sulfur are influenced by microbial processes that occur in the vast regions of the world's oceans. This is especially important for the sulfur cycle, where dimethylsulfide (DMS) release from algae can lead to changes in the acidity of the atmosphere and cloud formation.
15. Eutrophication can be caused by nutrient releases from rural and urban areas. Sources of nitrogen and phosphorus are particularly important in eutrophication. These mineral additions and the growth of photosynthetic microorganisms can lead to widespread hypoxia and anoxia in coastal and estuarine areas.
16. Lakes can be oligotrophic (low-nutrient) and eutrophic (high-nutrient) environments (**figure 29.17**). The addition of organic matter, especially with the microbial use and release of N and P, can lead to eutrophication.
17. In rivers the addition of organic matter can produce dissolved O₂ sag curves and diurnal O₂ changes in the latter stages of self-purification (**figure 29.19**). This occurs with materials added from point sources.
18. Much of the marine environment is covered by sea ice. Specially adapted microbial communities are found at the ice-water interface. Deep Antarctic lakes (such as Lake Vostok) and dry valley frozen lakes provide unique environments for microbes. In an environment such as Lake Vostok, the microorganisms may have been held in the ice for 420,000 years or more.
19. Water purification can involve the use of sedimentation, coagulation, chlorination, and rapid and slow sand filtration (**figure 29.20**). Chlorination may lead to the formation of organic disinfection by-products, including trihalomethane (THM) compounds, which are potential carcinogens.
20. *Cryptosporidium*, *Cyclospora*, viruses, and *Giardia* are of concern, as conventional water purification and chlorination will not always assure their removal and inactivation to acceptable limits.
21. Indicator organisms are used to indicate the presence of pathogenic microorganisms. Most probable number (MPN) and membrane filtration procedures are employed to estimate the number of indicator organisms present. Presence-absence (P-A) tests for coliforms and defined substrate tests for coliforms and *E. coli* allow 100 ml water volumes to be tested with minimum time and materials.
22. Molecular techniques based on the polymerase chain reaction (PCR) can be used to detect waterborne Shiga-toxin producing *E. coli* O157:H7 in 8 hours, when a preenrichment step is used.
23. The biochemical oxygen demand (BOD) test is an indirect measure of organic matter that can be oxidized by the aerobic microbial community. In this assay, oxygen should never limit the rate of reaction. The chemical oxygen demand (COD) and total organic carbon (TOC) tests provide information on carbon that is not biodegraded in the 5-day BOD test.
24. Conventional sewage treatment is a controlled intensification of natural self-purification processes, and it can involve primary, secondary, and tertiary treatment (**figure 29.27**).
25. Constructed wetlands involve the use of aquatic plants (floating, emergent, submerged) and their associated microorganisms for the treatment of liquid wastes (**figure 29.29**). These systems are being used in a wide range of environments. Domestic wastes, industrial effluents, and acid mine drainage (AMD) can be treated.
26. Home treatment systems operate on general self-purification principles. The septic tank (**figure 29.30**) provides anaerobic liquefaction and digestion whereas the aerobic leach field allows oxidation of the soluble effluent.
27. Groundwater is an important resource that can be affected by pollutants from septic tanks and other sources. This vital water source must be protected and improved.

Key Terms

activated sludge 659
anaerobic digestion 659
anoxic 635
barophile 644
biochemical oxygen demand (BOD) 657
bulking sludge 659
chemical oxygen demand (COD) 657
coagulation 652
coliform 654
constructed wetland 662
copiotrophic 638
defined substrate test 654
disinfection by-products (DBPs) 653
diurnal oxygen shift 650
epilimnion 648
eutrophic 648
eutrophication 648
extended aeration 659
fecal coliform 654
fecal enterococci 656
groundwater 663

high oxygen diffusion environment 635
hypolimnion 648
hypoxic 635
indicator organism 654
Ingoldian fungi 641
in situ treatment 663
low oxygen diffusion environment 635
membrane filtration technique 654
microbial loop 638
most probable number (MPN) 654
MUG 654
nanobacteria 640
nitrogen oxygen demand (NOD) 657
nonpoint source pollution 650
oligotrophic 648
ONPG 654
phytoplankton 638
point source pollution 650
potable 654
presence-absence (PA) test 654
primary treatment 658

rapid sand filter 652
Redfield ratio 638
reverse methanogenesis 645
secondary treatment 659
sedimentation basin 652
septic tank 663
settling basin 652
slow sand filter 653
sludge 658
tertiary treatment 661
thermocline 648
total organic carbon (TOC) 657
trickling filter 659
trihalomethanes (THMs) 653
ultramicrobacteria 640
virioplankton 643
wastewater treatment 658
wetlands 662
Winogradsky column 637

Questions for Thought and Review

1. Why might you describe the interaction of many microorganisms with oxygen as a "love-hate" relationship? How do these microorganisms deal with this important gas?
2. How might you measure the activity of nanobacteria or ultramicrobacteria, in spite of the fact that they are so difficult to culture and observe?
3. What are the functional strategies of organisms such as *Thioploca* and *Thiomargarita*? Where would you go to find similar organisms, and how would you search for them?
4. You wish to study the characteristics of microorganisms from a 10,000 meter depth in the ocean. What effects might decompression have on these organisms, and what type of equipment would you design to work with them under their natural conditions?
5. Discuss the advantages and disadvantages for microorganisms that become attached to surfaces.
6. With sea ice being such an important environment for specialized microorganisms, what might be the global-level effects of these microbial communities?
7. Why are levels of virus particles so high in many marine environments? Consider the requirements for viral replication. How old do

- you think these viral particles might be and why?
8. How might one manage waters to control *Pfiesteria piscicida*?
 9. How might it be possible to rejuvenate an aging eutrophic lake? Consider physical, chemical, and biological approaches.
 10. You wish to develop a constructed wetland for removal of metals from a stream at one site, and at another site, you wish to treat acid mine drainage. How might you approach each of these problems?
 11. What are the linkages between nutrient cycling in the oceans and soils?
 12. Why are indicator organisms still being used despite the fact that methods are available for the direct isolation of most pathogens that might occur in water?
 13. What alternatives, if any, can one use for protection against microbiological infection when swimming in polluted recreational water? Assume that you are part of a water rescue team.
 14. What are possible alternatives that could be used to eliminate N and P releases from sewage treatment systems? What suggestions could you make that might lead to new technologies?

Critical Thinking Questions

1. If we were able to decrease or increase the activity of the microbial loop in the ocean, which should be done and why? Would either shift be desirable?
2. There have been numerous "fish kills" in the waters of the Chesapeake Bay region recently. Fishermen blame poultry farmers (poultry waste disposal) and the overuse and subsequent runoff of residential fertilizers by homeowners. What components in poultry waste and fertilizer can create an ecological imbalance that would result in hundreds of dead fish floating in inlet waterways?

Additional Reading

The references provided at the end of chapter 28 also may be consulted for further information.

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